GPCR - Receptor Expression Evolution

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ABSTRACT

The purpose of this research was to better understand the expression patterns of monoamine receptors, in particular serotonin and dopamine receptors, in a single-cell sequencing dataset of three different insect species: Drosophila, Monomorium Pharaonis, and Aedes aegypti. We wanted to discover the distribution and diversity of monoamine receptor-expressing cells throughout the various cell clusters in the brain tissues of these insects, therefore we carried out single-cell sequencing on those tissues. We were able to identify and classify the cells that express serotonin and dopamine receptors through the use of sophisticated computational analysis, which provided us with important insights on the distribution of these receptors. The research showed that each of the three insect species exhibited a unique pattern of serotonin and dopamine receptor expression, respectively. There were certain trends in the expression profiles of these receptors across particular cell clusters; however, there were also some noticeable variations, which indicated that there were species-specific variations in the distribution of these receptors.The findings of this study contribute to our understanding of the neurobiological mechanisms involving monoamine receptors in insects. They also pave the way for further investigations into their roles in regulating behaviour and physiological processes, as well as potential implications for insect-related research, such as the management of vector-borne diseases and pest control.

1.INTRODUCTION

1.1 G-Coupled Receptor Protein

G-protein coupled receptors (GPCRs), are cell-surface receptors that play a crucial role in transducing extracellular signals into intracellular effector pathways through the activation of heterotrimeric G-proteins (Heng, Aubel and Fussenegger, 2013) GPCRs comprise the largest family of membrane proteins and are responsible for mediating most cellular responses to hormones and neurotransmitters. They serve as essential nodes in the communication network that connects the cell's interior with its external environment. Moreover, GPCRs are responsible for our sense of light, smell, and taste (Rosenbaum, Rasmussen and Kobilka, 2009)

The evolutionary significance of GPCRs becomes apparent when considering their presence in diverse organisms. They have been identified in the genomes of bacteria, yeast, plants, nematodes, and invertebrates, providing evidence for their ancient origin (Kroeze, Sheffler and Roth, 2003) This widespread distribution across different species suggests that GPCRs have played fundamental roles in cellular signaling throughout evolutionary history. GPCRs and G-protein signalling have been around for around 1.2 billion years, long before the divergence of land plants, fungi, and mammals(Schöneberg *et al.,* 2007)

The studies of GPCR genes in insects and their possible biological functions reveal that these genes may play a role in a wide variety of physiological processes, including reproduction, growth and development, stress responses, feeding pattern, behaviors and other physiological processes. (Liu, Wang, Li and Feng, 2021)

GPCR pathways are activated in insects by compounds like 20-hydroxyecdysone (20-E) or neuropeptides, which then control the expression of genes involved in processes like apoptosis and insect metamorphosis. (Liu, Wang, Li and Feng, 2021)The activation of the signaling pathway that is common to all members of this receptor family begins with the binding of the agonist to the unique surface receptor associated with that agonist. This binding event induces a conformational shift in the receptor, allowing it to engage

with the heterotrimeric G protein in the cell membrane, resulting in the formation of a high-affinity agonist-receptor-G protein complex. The interaction between the receptor and the G protein catalyzes the exchange of guanine nucleotides on a subunit of the G protein, ultimately leading to the separation of the G protein complex into its α and βγ subunits. The activated form of the G protein, with GTP attached to it, dissociates from the receptor and activates an effector protein, which modulates intracellular second messengers and transmits the signal (Strader *et al.,* 1994)

GPCRs exhibit an impressive capability to bind a diverse range of ligands, including hormones, neurotransmitters, ions, photons, odorants, and neuropeptides. These ligand-specific interactions lead to interactions with G proteins and arrestins, initiating downstream signaling pathways. These pathways include the adenylate cyclase/cyclic AMP (AC/cAMP), phosphatidylinositol/diacylglycerol/protein kinase C (PI/DAG/PKC), and voltage-gated calcium channel (Ca2+ channels) pathways, which are essential for growth, development, reproduction, and various physiological activities. (Liu *et al.,* 2021)

These receptors, commonly referred to as heptahelical receptors or 7-TM receptors, as they are characterized by a seven-transmembrane α-helical fold composed of seven membrane-spanning helices (Kroeze, Sheffler and Roth, 2003) They possess an extracellular N-terminus, which is exposed to the extracellular environment and often contains numerous glycosylation sites, as well as an intracellular C-terminus that faces the cytoplasm and provides access to potential phosphorylation sites. Ligand-specific binding occurs in the extracellular domains, while the intracellular regions interact with a heterotrimeric GTP-binding protein (Caers *et al.,* 2012). Analysis of the cloning and sequencing of more than a hundred members of this extensive receptor protein family has revealed that these receptors are distinguished by seven hydrophobic regions, each consisting of 20-25 amino acids. These regions are predicted to form transmembrane alpha helices and are interconnected by alternating extracellular and intracellular loops (Strader *et al.,* 1994)

The analysis uncovered the presence of 197 GPCRs encoded in Lucilia cuprina, encompassing various subtypes including rhodopsins (n=73), secretin receptors (n=18), metabotropic glutamate receptors (n=9), and unclassified proteins (Anstead *et al.,* 2015) Similarly, in the stick insect Carausius morosus, 430 putative GPCR transcripts were identified, with 43 of them exhibiting significant similarity to known GPCR sequences (Duan Şahbaz and Birgül Iyison, 2019) Additionally, 65 putative neuropeptide GPCR genes were discovered in spider mite, with 58 belonging to the A-family and 7 to the B-family (Veenstra, Rombauts and Grbić, 2012) Moreover, our investigation of the human proteome revealed that out of the 799 full-length GPCRs, 430 are primarily involved in sensory processing, such as smell, taste, and vision, making them less likely to be useful therapeutic targets.These GPCRs include the 8 opsin/rhodopsin-related receptors, 388 olfactory receptors, 6 vomeronasal-like receptors, 25 bitter-taste receptors (T2Rs) and 3 sweet/umami taste receptors Conversely, the remaining 369 GPCRs hold great promise as potential therapeutic targets for a wide range of human diseases. (Heng, Aubel and Fussenegger, 2013)

GPCR have been the subject of much research due to their pathophysiology and pharmacological significance. 475 GPCR-targeting medicines with a current FDA-approval status were found. This amounts to around 34% of all medications authorized by the FDA. The actions of the 475 authorized medications are mediated by a total of 108 GPCR targets, which together account for 27% of the human non-olfactory GPCRs. 314 of the currently available medicines are designed to act on aminergic receptors, which are recognized as effective therapeutic targets. (Hauser *et al.,* 2017)

Data reveals that the indications for GPCR-targeted medicines are growing, moving beyond the traditional domains of hypertension, allergies, analgesics, schizophrenia, and depression, and into other areas including Alzheimer's disease and obesity. Multiple sclerosis (MS), smoking cessation, short bowel syndrome, and hypocalcaemia are only some of the newer diseases for which GPCRs have been addressed. (Hauser *et al.,* 2017) Drugs that either stimulate (agonists) or block (antagonists) GPCRs are critical in managing diseases affecting major organs, including the brain, heart, lungs, metabolic

and urinary systems. For example, AT1 angiotensin II receptor blockers are used to reduce kidney damage from diabetes, manage hypertension, and treat heart failure. Alpha-1 adrenergic receptor blockers are utilized to manage symptoms and slow progression of benign prostatic hyperplasia. Beta adrenergic blockers, which act on beta-1 and/or beta-2 receptors, are vital in improving survival rates in heart failure patients and are also employed in treating hypertension and coronary heart disease. In contrast, beta-2 adrenergic agonists are central in managing asthma and chronic obstructive pulmonary disease and delaying premature labor. Drugs that block dopamine receptors, specifically D2, are standard treatments for schizophrenia, while dopamine precursors like levodopa, which boost dopamine levels, are extensively used for Parkinson's disease. Lastly, medications that inhibit serotonin uptake indirectly stimulate various serotonin receptors, aiding in treating depression, and possibly contributing to the effects of certain antipsychotic drugs. (Insel *et al.,* 2007)

Classification of GPCRs in organisms is often done using one of two systems: the A-F system or the GRAFS framework. G protein-coupled receptors (GPCRs) may be broken down into six groups, designated A through F, based on their structural and functional similarities. The largest class of GPCRs is A, the rhodopsin-like family; B, the secretin receptor family; C, the metabotropic glutamate family; D, the fungal mating pheromone receptors; E, all the cAMP receptors; F, the frizzled/smoothened receptors. (Liu, Li *et al.,* 2021) There are five major families in the GRAFS system: glutamate (G), rhodopsin (R), adhesion (A), frizzled/taste2 (F), and secretin (S). In addition, the extensive rhodopsin family (R/class A) is classified into 13 distinct groupings. Grouping was done using stringent phylogenetic criteria, and the few human receptors that did not cluster were put into a new category termed as "Other 7TM receptors." (Schiöth and Fredriksson, 2005)

The typical function of GPCRs is to relate the binding of agonists to the activation by particular heterotrimeric G proteins, which ultimately results in the control of downstream effector proteins.(Rosenbaum, Rasmussen and Kobilka, 2009)

1.2 Single cell sequencing

The sequencing of nucleic acids within single cells, also known as single-cell sequencing (scSeq), has led to a revolution in our knowledge of the diversity of cell types at different levels of the biological hierarchy (for example, tissue, organ, individual), as well as an advancement in our understanding of the relationships between genotype and phenotype. (Alfieri *et al.,* 2022)

Several investigations based on cell separation and individual genomic library preparation were reported after the initial scRNA-seq experiment in a four-cell-stage blastomere. (González-Silva, Quevedo and Varela, 2020)

Using single-cell RNA sequencing (scRNA-seq) technologies, RNA-seq may be done on individual cells, allowing for the study of cell-specific changes in RNA expression. Accordingly, unlike conventional RNA-seq, scRNA-seq permits statistical studies that may provide additional biological insights. (Yip, Sham and Wang, 2019)

In order to perform mRNA sequencing from a single cell (scRNA-seq), two specific challenges that aren't prevalent in conventional population-level techniques need to be tackled: (1) the isolation of individual cells and (2) the magnification of extremely small quantities of mRNA from a cell. The procedure for every scRNA-seq study adheres to a common fundamental approach. Initially, an individual cell is isolated and broken down, followed by a process called reverse transcription, which helps in selecting mRNA (via poly[T] priming) and generating cDNA. The very small quantities of cDNA produced are then magnified using either PCR or in vitro transcription. Eventually, the magnified cDNA is utilized for the preparation of the sequencing library. (Kolodziejczyk *et al.,* 2015)

2 Methods

2.1 Systematic GPCR Sequence Discovery and Orthogroup Clustering

To discover probable G-protein coupled receptor (GPCR) sequences, a systematic study of entire proteomes from diverse animal species was done. Phobius was used for the identification, which allowed for the accurate discovery of sequences having precisely 7 transmembrane domains, as is typical of GPCRs. The GPCR sequences were then clustered into orthogroups using OrthoFinder, which groups related sequences based on evolutionary ties. Thus, the evolutionary distribution and conservation of GPCRs across many animal species could be studied.Seven transmembrane proteins and their respective ortho groups were listed as output.This was all preparatory work performed in the laboratory.

2.2 Single cell sequencing

The single cell sequencing was performed on Data set obtained from previous studies

The aedes data set was obtained from study conducted on male and female Aedes aegypti Brain (Cui, Behura and Franz, 2022)

The Drosophila data set was obtained from study conducted on the Drosophila Brain cells (Davie *et al.,* 2018)

The Monomorium Pharaonis data set was obtained from the study on the ant super organism (Li *et al.,* 2022)

The Seurat tool in RStudio was used to analyse scRNA-seq data from three species (Monomorium Pharaonis,Drosophila and Aedes aegypti) in this study. First, the data was imported into R and plotted it using violin plots, which show both feature counts and mRNA counts. To remove poor-quality cells from the dataset, this visualisation was employed.

Unwanted cells were eliminated, and then the data was normalised. Next, the highly variable characteristics identified highly to capture the most informative genes

contributing to cell-to-cell variance and normalised the data to standardise the expression of each gene.

After that, principal component analysis (PCA) was carried out on the scaled data in order to minimise the dimensionality of the dataset and locate the principal components (PCs) that explain the most variance in the data. To determine the optimal number of top PCs to use, a jackstraw plot was generated.The jackstraw plot helps identify significant PCs by comparing the actual variance explained by each PC to that of randomly permuted data.

The UMAP (Uniform Manifold Approximation and Projection) technique was used to visualise the cells in a low-dimensional space based on the data from the jackstraw plot, which aided in the detection of cell clusters.

Marker genes were gathered to further characterise each cell cluster. The markers were narrowed down to those that indicated genes with significantly differing expression levels between the clusters by setting the average log2 fold change (avg_log2fc) criterion to be greater than 1.

The next phase in the study was to annotate the cell clusters with biological meaning once the marker genes for each cluster had been determined.In order to classify the cell clusters, The cell clusters are annotated using known marker genes that have been previously characterised and linked to particular cell types.

Dot plots are used to visualise the expression patterns of the genes identified as significant marker genes for each species.. The expression of these markers in various cell clusters within a species can be visualised using a dot plot, which gives a useful visual representation of this data. The dot plot is created by showing distinct cell clusters along the x-axis, and plotting the chosen marker genes along the y-axis. The expression of each marker gene is shown along the y-axis, with each dot in the plot representing a cell cluster. Larger dots denote cells with more intense expression of a marker gene, which may indicate enrichment or unique expression of that gene in certain cell groups.

2.3 Aedes Aegypti Brain

The data was collected from a previous study conducted on aedes aegypti Brain cells.(Cui, Behura and Franz, 2022) carried out a comprehensive study on the single-cell RNA sequencing (scRNA-seq) data from Aedes aegypti. The information on male and female species was gathered independently for this study. It comprised roughly between 3,000 to 4,000 cells derived from male and female individuals. To get rid of the cells that were of poor quality the data set was filtered. Only the specimens having a feature count of between 200 and 2500 were chosen from the male Aedes aegypti data. In the female data set, the individuals that met the selection criteria of having a feature count between 400 and 2800 and an RNA count of more than 400 were chosen.

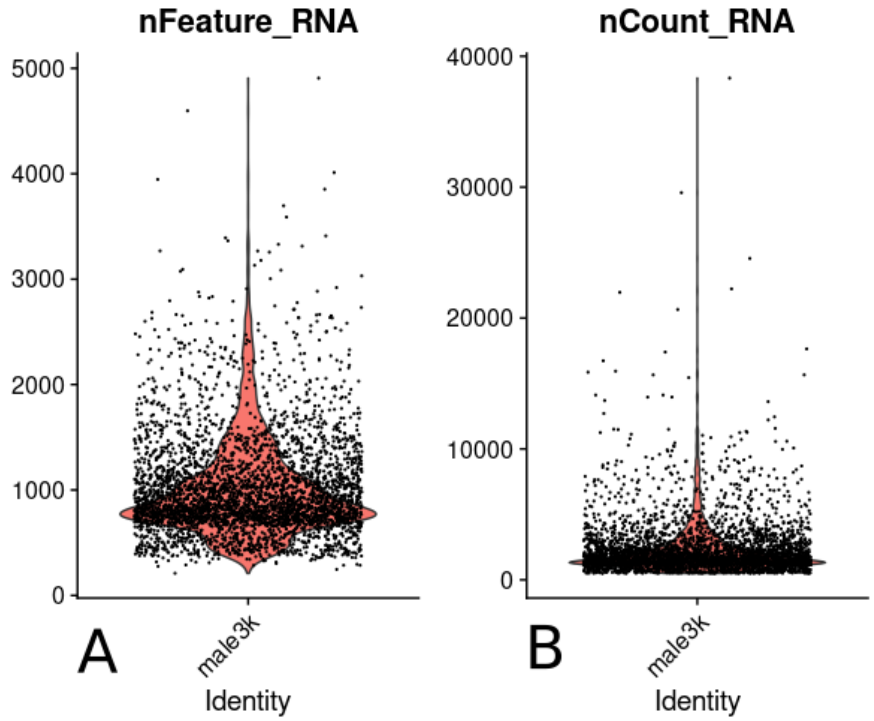


Figure 1. Violin plot of the male Aedes aegypti.In figure A the y axis is the number of feature in the data set and in Figure B y axis represent the mRNA count in the dataset of aedes aegypti. Most of the cells had a feature count around 200-1500 and mRNA count less than 500

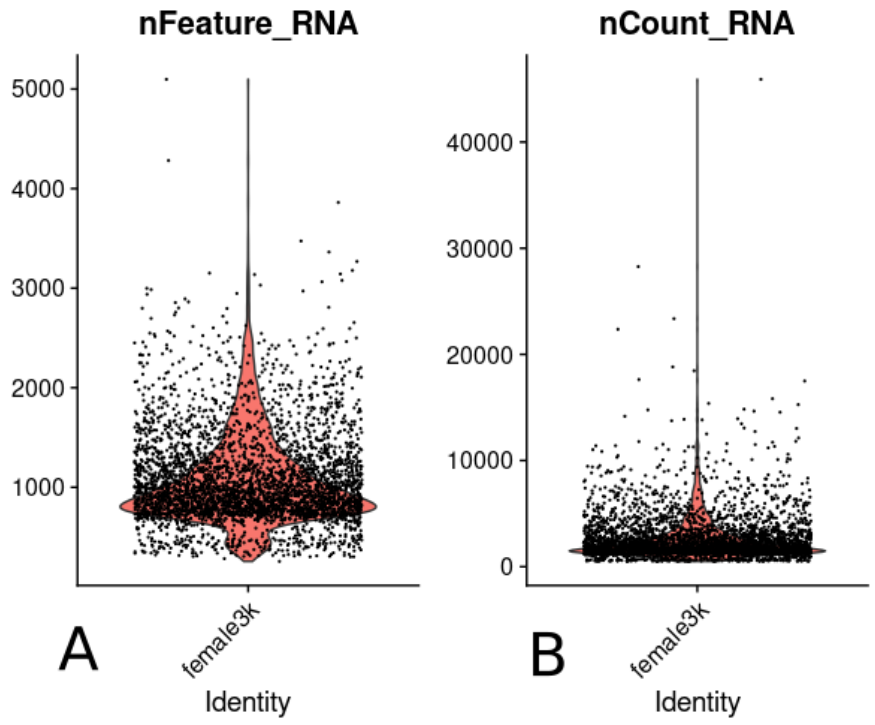


Figure 2.Violin plot of the female Aedes aegypti dataset In figure A the y axis represent the number of features and in figure B the y axis represent the mRNA count. Most of the cells had a feature count in the range of 400-2800 and had the mRNA count less than 400

The male and female dataset were combined after being normalized so that undesirable variance could be eliminated. 7 450 cells were a part of the combined data set. After that, a principal component analysis is carried out in order to determine which aspect of the data set is the most significant. I choose to construct a heat map of the principal component (PC).

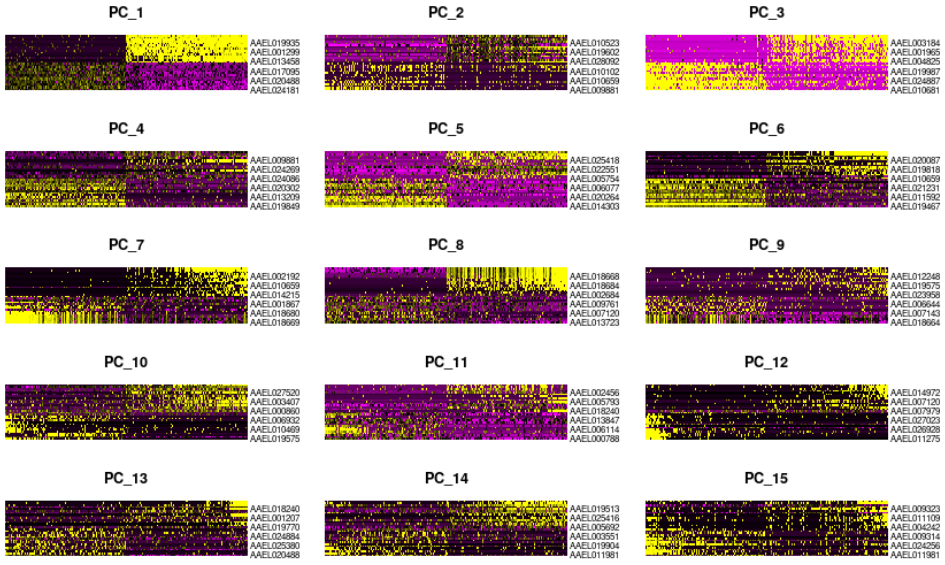


Figure 3 Heat map of the principal component present in the merged data. The y axis represent the annotation of the cell

For identifying the important PC I choose to create a Jackstraw plot.

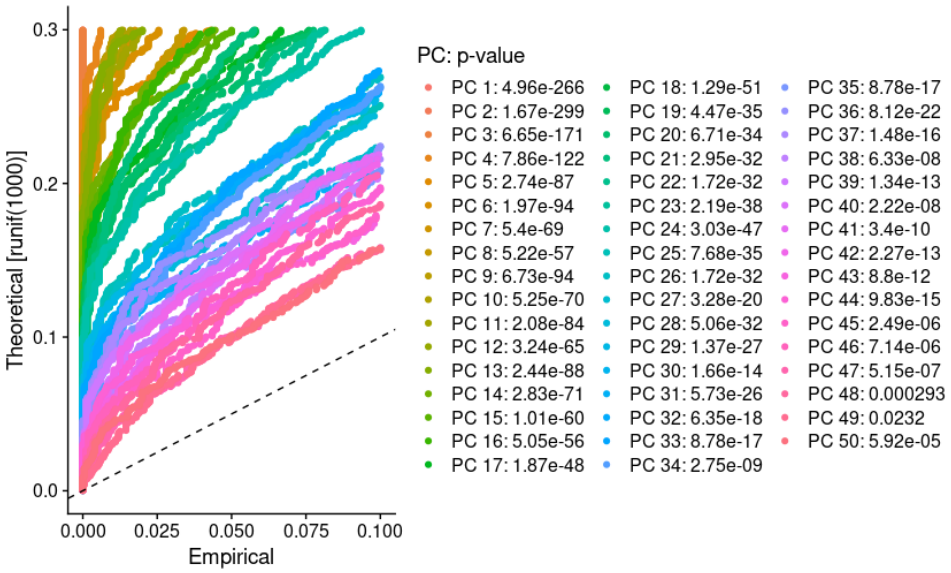


Figure 4 The jackstraw plot of the principal component of the dataset.It is evident from the figure that there are 50 significant principal components.

Based on the information provided by the jack straw plot of the PC,there was 50 significant principal component so,i choose that the appropriate dimension for the clustering of the complete data set would be 1:50, and this was accomplished with a resolution of 0.5. Approximately 31 clusters were included in the Aedes data set. From among these clusters, the Custer 0 and 1 cluster seems to be significantly large cluster comparatively.

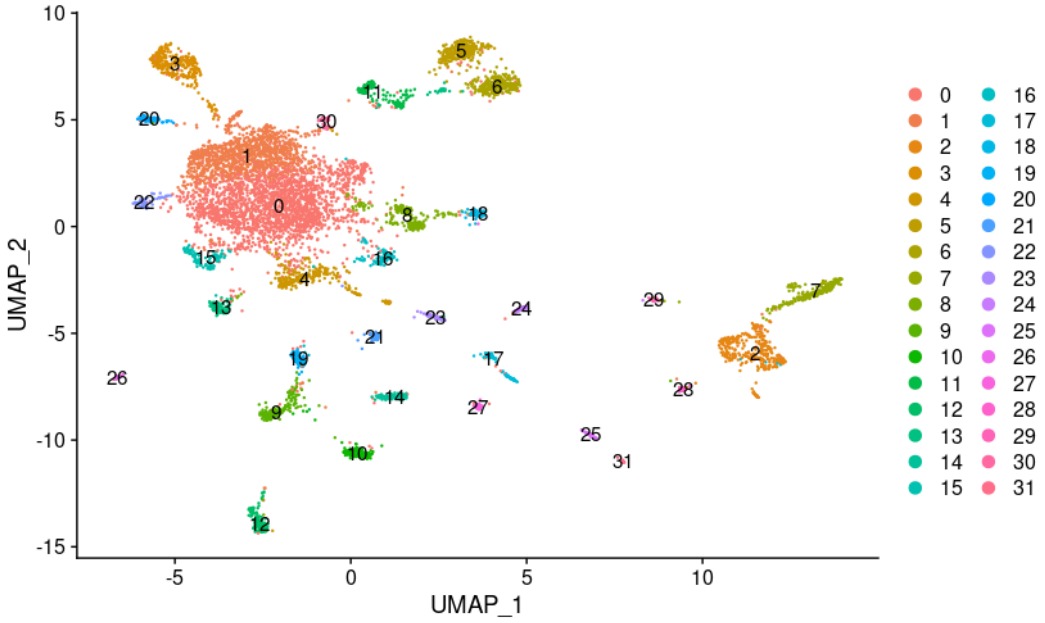


Figure 5 The UMAP of the Aedes data with a resolution of 0.5.It produced 31 significant clusters.

We then decided to compile a list of marker genes specific to each cluster. There were a large number of marker genes for each cluster, and from this list of marker genes, the ones that had an average log2FC greater than 1 were selected. Using this list of maker genes, we tried to determine the cell type that corresponds to each cluster. In order to gain information on the marker genes, we looked at past research. After that, we annotated the cluster using the marker genes that were significant in each individual cluster. We located the maker genes, used a dot plot to examine the presence of the selected marker genes in the cluster, and then determined which of the possible matches was the most appropriate. We have annotated the cluster based on marker genes for the cell type.

Annotating the Aedes Aegypti cluster

We able to annotate 13 clusters to cell types based on information from previous researches on the Aedes aegypi Brain cells (Cui, Behura and Franz, 2022)

Cluster 29 was initially annotated as astrocytes, which is a type of glia cells due to the presence of markers reversed polarity (repo) , wunen-2(wun2) and sprouty(sty) .But on the later stage of our analysis this cluster was named as glia cells so that we could have a common cell type across species and compare them with the clusters of other species.

Cell type	**Markers in *Ae. aegypti***	**Markers in *Drosophila***
astrocyte	AAEL027131, AAEL007322, AAEL014510	repo, wun2, sty
glia	AAEL027131, AAEL010145	repo, nrv2
KC	AAEL002321, AAEL005834, AAEL019691	eyeless, DopR2, sNPF
Mi	AAEL011643, AAEL007221	hth, bsh

monoaminergic neuron	AAEL026276, AAEL024732, AAEL005581	*Vmat, DAT, SerT*
Pm	AAEL011643, AAEL007120	
PN	AAEL019970, AAEL005507	ct, acj6
vPN	AAEL019970, AAEL005507, AAEL019457	*ct, acj6, Lim 1*

Table 1.The marker genes used for the annotation of the Aedes aegypti.

CLUSTER	Cell Type	Annoatation
2	glia 1	glia
3	KC 1	KC

5	vPN 1	PN
6	vPN 2	PN
7	glia 2	glia
10	Mi1	Mi1
11	PN 1	PN
13	KC 2	KC
16	monoaminergic neuron	monoaminergic
18	KC 3	KC
22	Pm	Pm
28	glia 3	glia

29	astrocyte	glia

Table 2. The Clsuter that have been annotated in Aedes.The cluster number of data is mentioned along with the cell type and our annotation.

We next compared the list of complete genes included in all clusters to the 7 transmembrane list we created to filter out those having 7 transmembrane structure.From the list of ortho groups of genes, we attempted to locate the ortho group of the filtered list of seven transmembrane structures in our database. Only those in ortho group 6 were chosen from this list since GPCRs are present in this ortho group. Because serotonin and dopamine were of special interest to us, we looked for the drosophila homologues of serotonin and dopamine receptors in Aedes.In order to accomplish this, a software called FigTree, which is used for visualising and analysing phylogenetic trees, was utilised to visualise the phylogenetic tree of these receptors. The homologous receptors for serotonin and dopamine were located throughout the tree, demonstrating that Aedes contains both of these neurotransmitters.We compiled a list of the serotonin receptor as well as the dopamine receptor in Aedes aegypti

DROSOPHILIA	AEDES AEGYPTI
5-HT1 A	AAEL08360
5-HT1 B	AAEL017272

5-HT7	AAEL025125 AAEL027242
5-HT2 A	AAEL019804

Table 3 The serotonin receptor in Aedes aegypti.These are the aedes serotonin receptor that is homologous to serotonin receptor in drosophila

DROSOPHILIA	AEDES AEGYPTI
DOP 1R1	AAEL019437
DOP 1R2	AAEL005834
DOP EcR	AAEL00266
DOP 2R	AAEL019766

Table 4 . The Dopamine receptor in Aedes that is homologous to the dopamine receptor in Drosophila.

Dot plots were used in order to ascertain whether or not clusters contained serotonin or dopamine recptor.In addition, both their precantage as well as their average expression was observed.

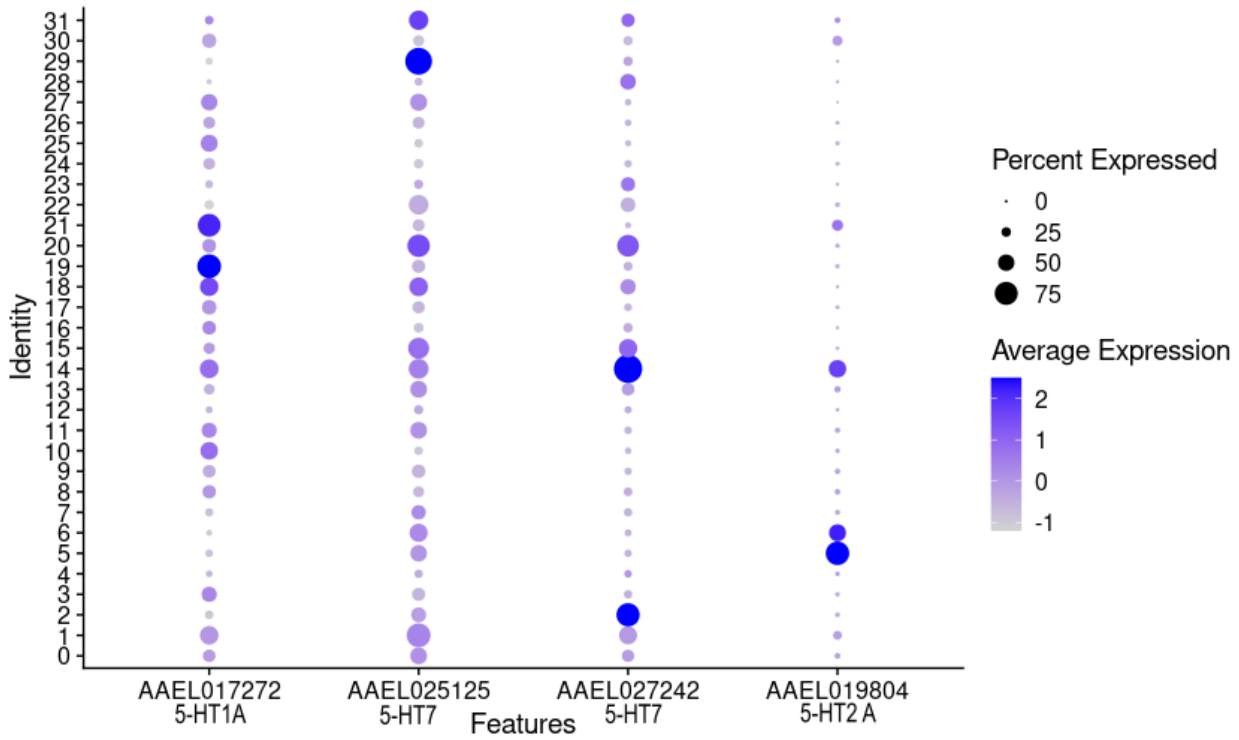


Figure 6.Serotonin receptors presence in cluster of Aedes aegypti

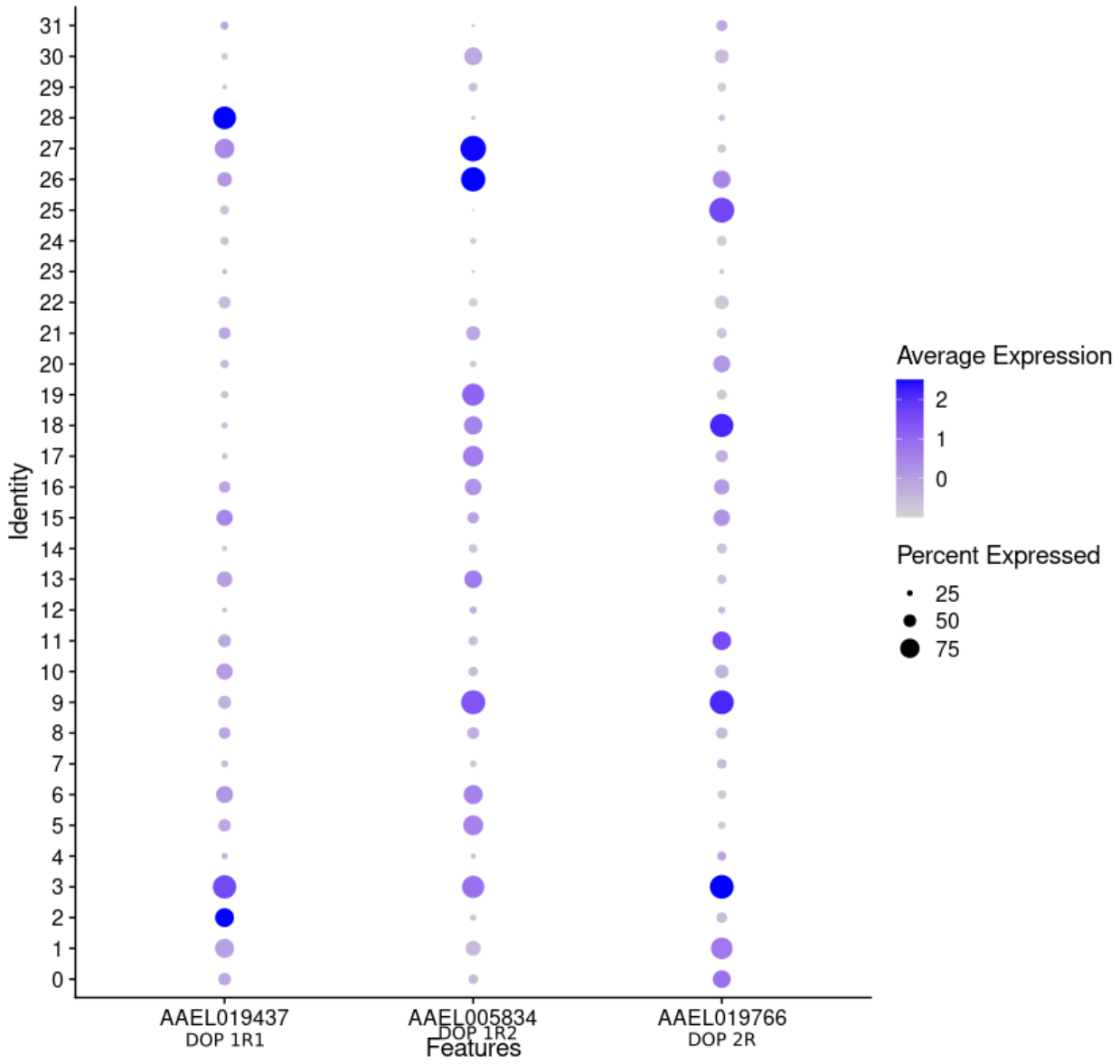


Figure 7.Dopamine receptors presence in cluster of Aedes aegypti

2.4 DROSOPHILA BRAIN

The Drosophila Brain dataset was obtained from previous study conducted on Drosophila Brain.(Davie *et al.,* 2018) Approximately 40,000 cells made up the first part of the dataset obtained from Drosophila.A filtering procedure was carried out in order to concentrate more on the genes that are active and important.Only those individuals with expression levels ranging from 200 to 2500 were chosen for analysis.Following the application of filters, the data set included 38000 cells.

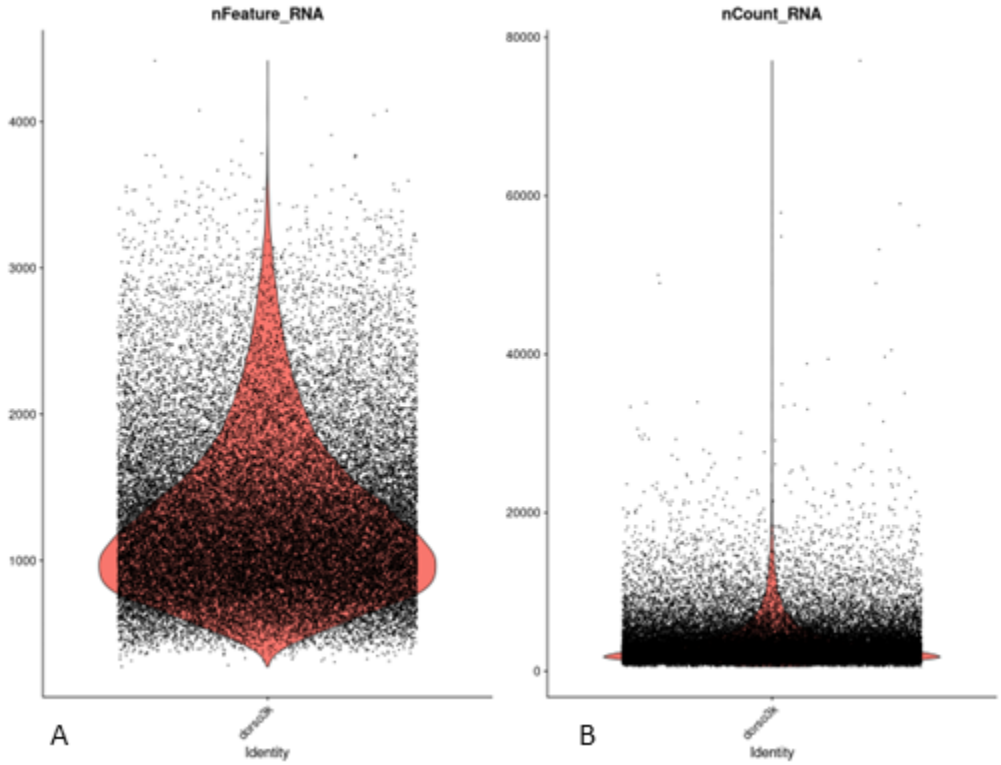


Figure 8 The violin plot of the Drosophila Brain dataset.In the figure A represent the feature count of the dataset.It has a peak around 1000 and B represent the mRNA count in the data .Most of the cells have a range around 200-1000.

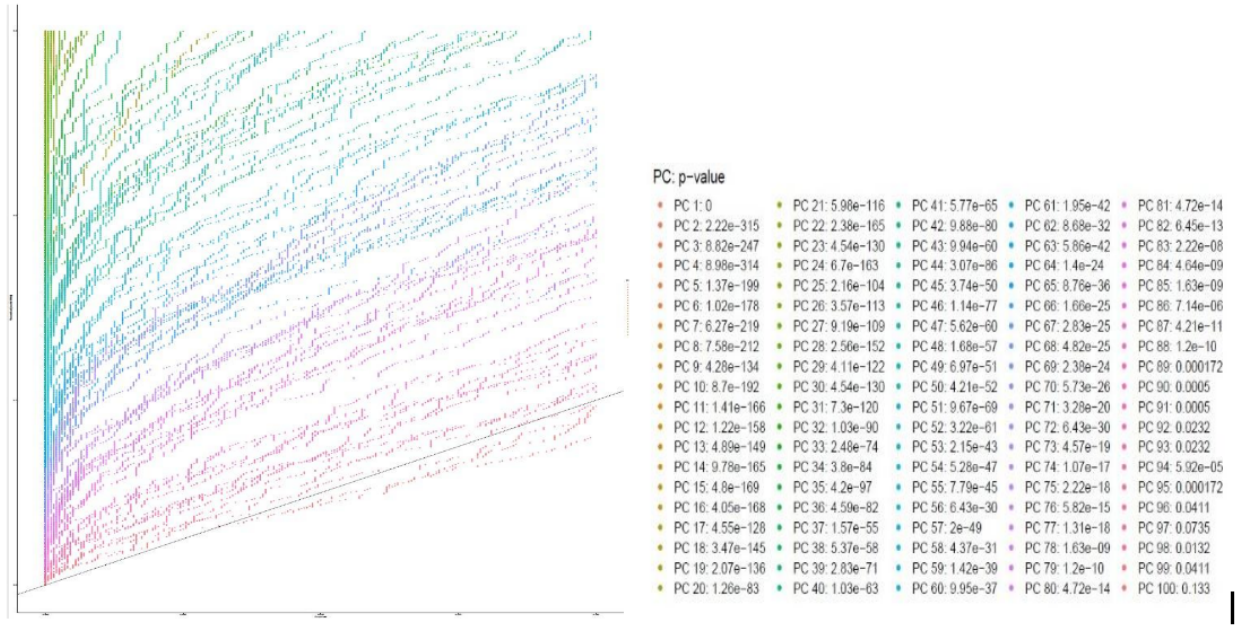


Figure 9 The jackstraw plot of the Drosophila Brain dataset .It is clear that the principal component from 1-100 seems to to significant.

The Jackstraw plot indicated that there were one hundred significantly important principal components.The dataset was clustered using a resolution of 0.5, which resulted in 67 groups indicating groupings of cells with similar patterns of gene expression. The clustered sample indicated that out of the total of 67 clusters, Clusters 0 and 1 were the biggest, followed by Cluster 2. This demonstrates that the number of cells in these three clusters is much larger than the number of cells in the other clusters.

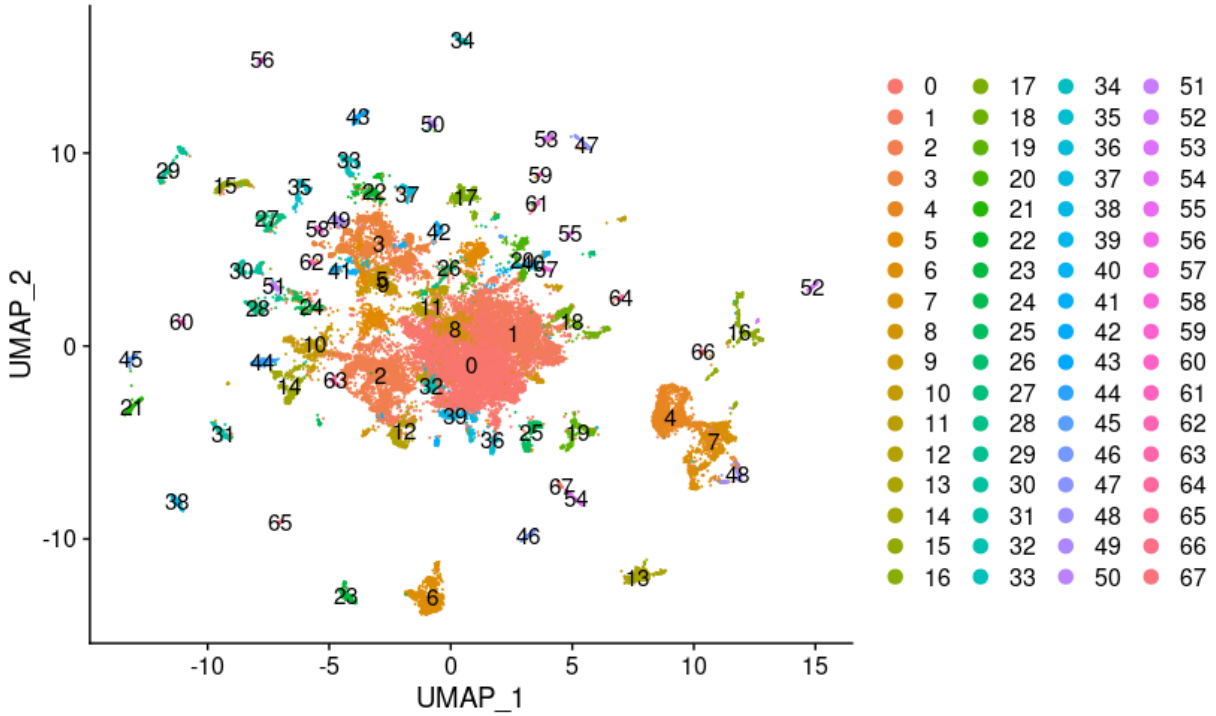


Figure 10. The UMAP of the Drosophila Brain.It produced 67 clusters

After collecting the markers from these clusters, we narrowed our choices down to those that had an avg_log2FC value that was more than one and tried annotating the clusters based on the marker genes present in each clusters.After determining the cell type that each cluster corresponds to based on the marker genes, we labeled them appropriately. Additionally, we searched for the Drosophila markers that were used to annotate Aedes clusters and utilized that data as well for the annotation process.

We choose to name the cluster based on cell type group rather than cell sub type which we did initially so that, it would be easier for the cross-speceis comparison. Instead of labeling the clusters separately as surface glia, cortical glia, and astrocytes, we simply referred to all of these clusters together as glia since they are composed entirely of glia cells.In order to verify this, we decided to utilize the glia drosophila markers that were used to annotate the Aedes aegypti clusters.

We did the same thing with Kenyon cells, referring to all of them collectively as KC cells rather than by their previously-defined subtypes of alpha, beta, and gamma. This made it much easier to examine how the clusters were across various species.

Cell Type	MARKERS IN DROSOPHILIA	Markers used for Aedes	Cluster Name
Surafce glia	Indy	repo,nrv2	glia
CTX (Cortex glia)	wrapper	repo,nrv2	glia
AST(glia)	alrm	repo,nrv2	glia
Gamma KC	sNPF,trio	ey,trio,sNPF,DopR2	KC
alpha/beta KC	sNPF	ey,trio,sNPF,DopR2	KC
alpha'/beta' KC	trio	ey,trio,sNPF,DopR2	KC
Monoaminergic	Vmat,DAT	ddc	Monoaminergic
Olfactory Projectial Neuron	tup,otp,c15,hth		PN

DAL	dally		DAL
PEP	pdf,dimm		PEP
Mi1	hth,bsh		Mi1
Mip	Mip		Mip
T1	Eaat1		T1
Hemocytes	Hml		Hemocytes
OCTY	Tdc2		OCTY
Pm	Lim3		Pm

Photoreceptors	ort		Photoreceptors
Pm3	Vsx1		Pm3

Table 5.The markers used for annotation of the Drosophila Brain clusters.The Drosophila markers which was used to annotate the Aedes was also used to annotate the Drosophila.

CLUSTER ID	Cell type	Annoatation
6	Gamma KC	KC
7	dally	Dally
13	alpha/beta KC	KC
16	SURFACE GLIA	glia

18	PEP	PEP
19	Monoaminergic	monoaminergic neurons
20	Mi1	Mi1
23	alpha’/beta’ KC	KC
25	Mip	Mip
32	OPN	PN
45	T1	T1
48	CTX CORTEX GLIA (glia)	glia
52	Hemocytes	Hemocytes
54	OCTY	OCTY

59	Pm	Pm
61	Photoreceptors	Photoreceptors
65	Pm3	Pm3
66	AST (glia)	glia

Table 6. The annotation of the clusters in Drosphila Brain.The first column is the cluster number of our data set.

A dot plot was created for the purpose of identifying the cluster in which the markers are present, and the clusters were then annotated accordingly.

Later, we compiled a list of all the cells that were included in the data set, compared it to the seven transmembrane lists that we previously had, and filtered the results based on the criteria "avg_log2FC > 1”.Moreover, the ortho group of the seven transmembrane cells that are included in our dataset was discovered.Only the cells in ortho group 6 were selected from this list for further study.

Serotonin Receptor
5-HT1 A
5-HT1 B
5-HT 7
5-HT 2A

Table 7 The serotonin Receptor in Drosophila

Dopamine Recptor
Dop 1R1
Dop 1R2
Dop EcR
Dop 2R

Table 8 The Dopamine Receptor in Drosophila

The serotonin and dopamine (HT1A, HT1B, HT 2, HT 7,DOP 1R1, DOP1R2, DOPEcR, DOP 2R) in which we are interested were the focal point of our attention, and a dot plot was utilised to identify the cluster in which these receptors were expressed. The expression percentage of these receptors was also recorded. Only the cluster with a pct.exp more than 25 and an avg.exp.scaled greater than 0.5 caught our attention.

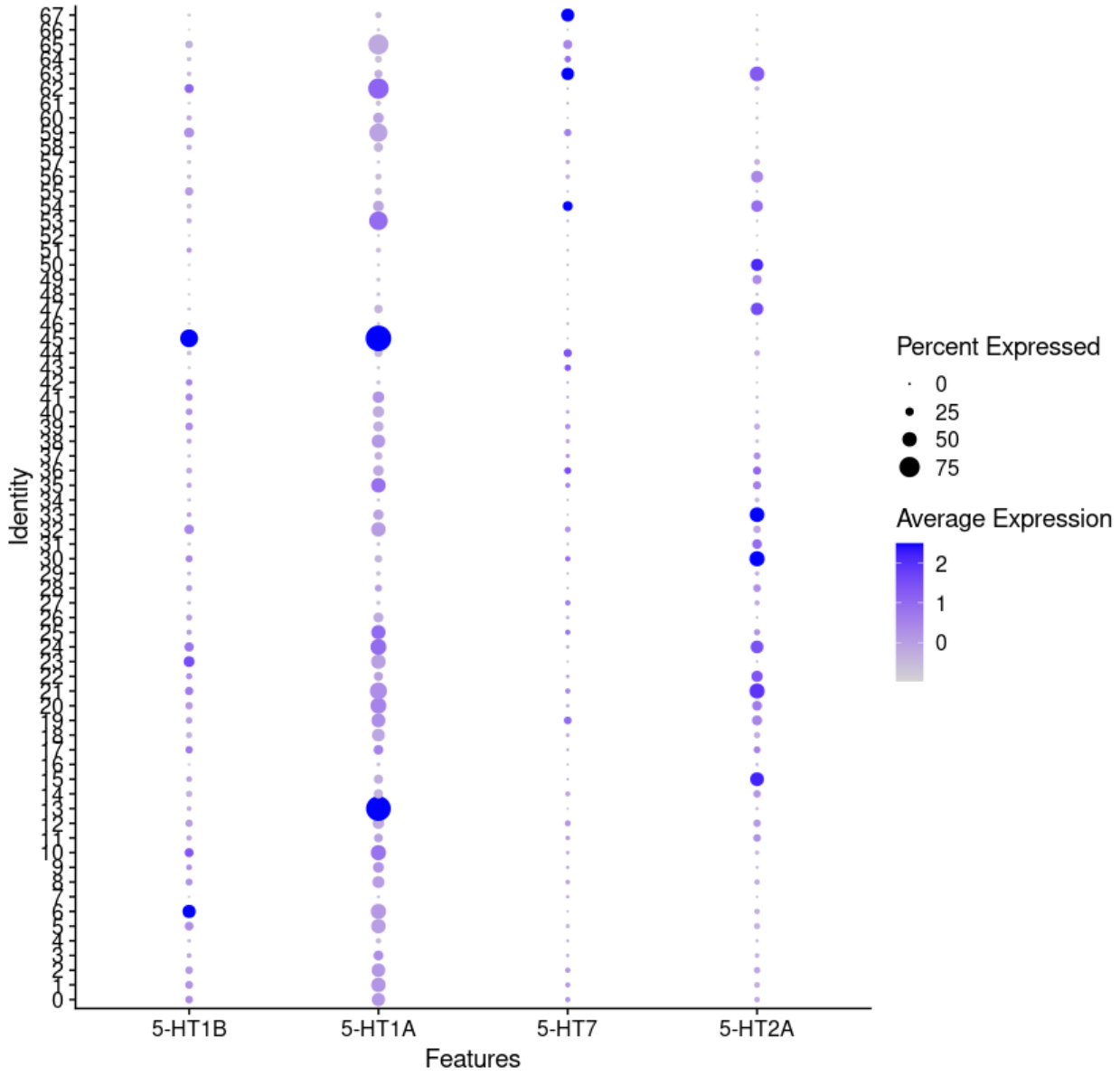


Figure 11 The serotonin receptor presence in the custer of the drosophila brain.

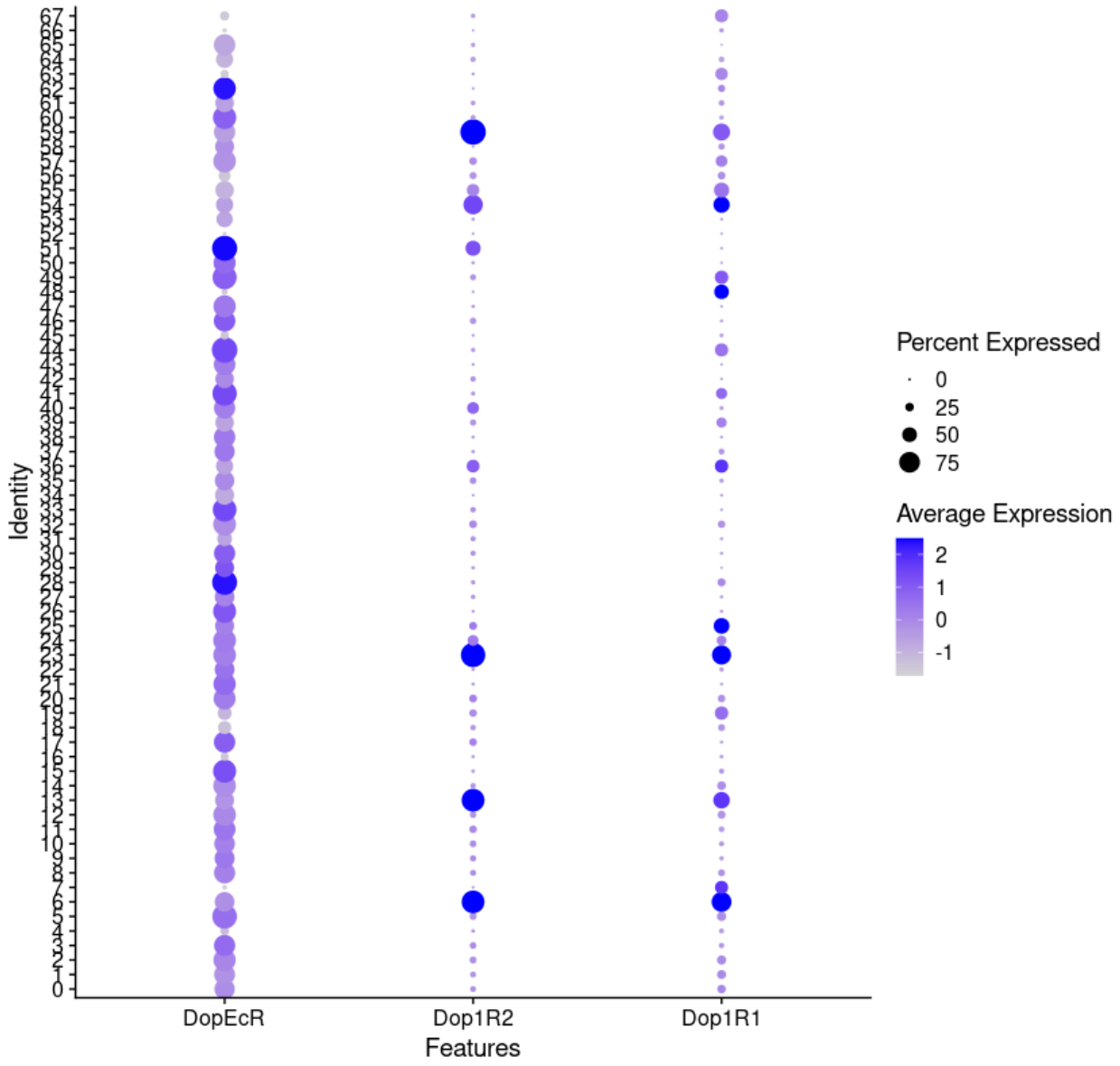


Figure 12 The Dopamine receptor presence in the cluster of the Drosophila Brain

2.5 Monomorium Pharaonis Brain

The Monomorium Pharaonis Brain data set was obtained from previous study conducted on the species.(Li *et al.,* 2022)Due to computational limitations, we had to reduce the data set to a manageable subsample of 25000 significant cells.In the first phase of data preparation, we restricted our analysis to genes with modest expression levels (between 200 and 1000 reads per gene). This filtering was performed with the intention of boosting the quality of subsequent studies by concentrating on physiologically significant genes and decreasing background noise. After filtering, the data was normalised to guarantee that detected variations in gene expression were biologically significant.

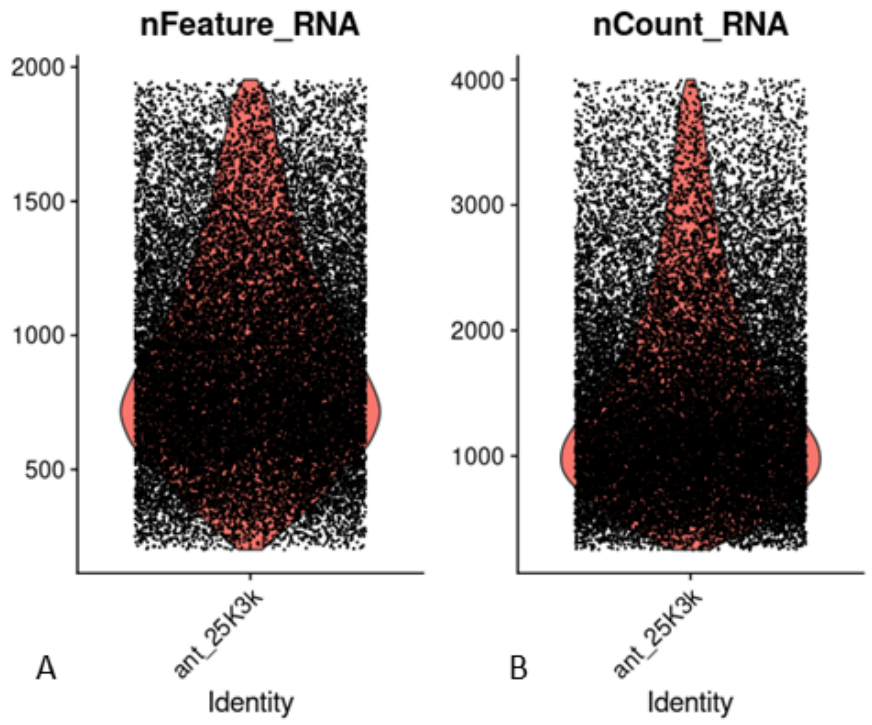


Figure 13 The violin Plot of the Monomorium Pharaonis Dataset.The figure A represnt the feature count in the dataset and figure B represent the mRNA count in the dataset.

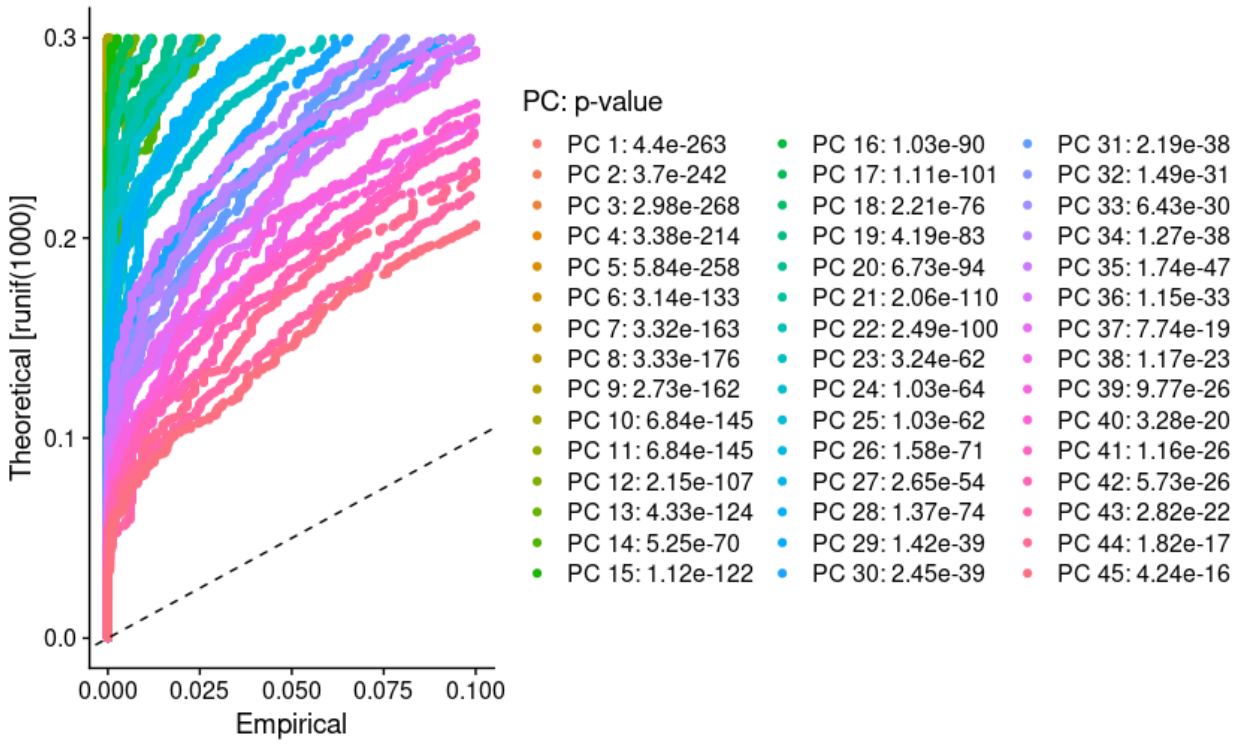


Figure 14 The jackstraw plot of the Monomorium Pharaonis dataset .It is evident from the plot that the principal component from 1-45 is significant

We used dimensionality reduction with the jackstraw plot to look further into the cellular heterogeneity seen in Monomorium Pharaonis. Dimensions 1-45 were picked from the plot because they represented the most important sources of variance in the data. This improved the efficiency of subsequent studies and the visualisation of the cellular landscape by decreasing their size.

To further separate cells with comparable gene expression patterns, a clustering analysis was performed at a resolution of 0.5. Serotonergic and dopaminergic cells are well-known for their functions in neurotransmission and behaviour, yet at this resolution, no substantial clusters associated with these cells were seen. This indicated that a higher resolution was required to provide a better and more complete view of the various cell types in the ant dataset.

Increasing the resolution to 2.5 allowed us to see more detail inside individual cells, with a total of 47 clusters visible in comparison to the original 37 clusters seen at the lesser resolution. By dividing larger clusters into smaller, unique subpopulations, made possible by the improved resolution, we were able to discover finer cellular diversity within Monomorium Pharaonis.

Later we generated a list of marker genes for each cluster. We selected only the marker genes that showed substantial differential expression across clusters based on an average log2 value larger than one. The default annotation for the marker genes was labelled as "Pharaoh-ant-*****," which lacked clarity and made it difficult to discern the functional importance of these genes in the context of the ant species that was being studied. To solve this problem, we looked at the study paper from which this data is taken to learn more about these marker genes. Research on Monomorium Pharaonis used Drosophila analogues to determine the marker genes for the ant species.we looked for the then got a better understanding of the Markers in Monomorium.We checked to see if those markers were present in each cluster of Monomorium Pharaonis using a Dot plot. This allowed us to annotate the clusters according to the unique markers they contained.

Using this method, we were able to gain a deeper understanding of the marker genes' meaning and the potential identities of the many cell clusters found in Monomorium Pharaonis. We were able to better comprehend the cellular heterogeneity found in the ant species by linking certain markers with distinct cell types or subpopulations.

Marker in Drosophila	**GENE Name in ANT**	**Cell Type**	**Cluster annotation**
Pka-C1	Pharaoh-ant-17936	Kenyon cells	KC

CaMK||	Pharaoh-ant-13966	Kenyon cells	KC
trio	Pharaoh-ant-01234	Kenyon cells	KC
zyd	Pharaoh-ant-06337	Cotex glia	glia
wrapper	Pharaoh-ant-07329	Cotex glia	glia
Indy	Pharaoh-ant-04520	Ensheathing glia	glia
vKg	Pharaoh-ant-04838	Ensheathing glia	glia
Mdr49	Pharaoh-ant-00624	Ensheathing glia	glia
Idgf4	Pharaoh-ant-17376	Surface glia	glia
CG8745	Pharaoh-ant-13676	Surface glia	glia
Tsf1	Pharaoh-ant-07850	Surface glia	glia

Gat	Pharaoh-ant-11362	Astrocytes	glia
GS1	Pharaoh-ant-17185	Astrocytes	glia
Ddc	Pharaoh-ant-04831	Monoaminergic neurons	Monoaminergic neurons
DAT	Pharaoh-ant-17489	Monoaminergic neurons	Monoaminergic neurons
Vmat	Pharaoh-ant-11830	Monoaminergic neurons	Monoaminergic neurons
acj6	Pharaoh-ant-10873	Olfactory projectional neuron	PN
trp	Pharaoh-ant-06879		Photoreceptors
ninaC	Pharaoh-ant-11119		Photoreceptors

ninaB	Pharaoh-ant-13171		Photoreceptors
Drgx	Pharaoh-ant-13626	Optic lobe T4/T5	Optic lobe T4/T5

Table 9. The markers used for the annotation of the Monomorium Pharaonis

CLUSTER	Cell Type	Annotation
4	Kenyon cells	KC
11	Kenyon cells	KC
12	Kenyon cells	KC
13	Kenyon cells	KC
14	Olfactory projectional neuron	PN
18	Kenyon cells	KC

19	Optic lobe T4/T5	Optic lobe T4/T5
21	Astrocytes	glia
24	Kenyon cells	KC
27	Monoaminergic neurons	Monoaminergic
29	Surface glia	glia
30	Ensheathing glia	glia
33	Kenyon cells	KC
40	Cotex glia	glia

Table 10 .The cell clusters of Monomorium Pharonis data set that have been annotated.

The scRNA-seq data from Monomorium Pharaonis was subjected to further screening. For this purpose, cells expressing genes encoding seven-transmembrane proteins were prioritized, as GPCR are themselves seven-transmembrane receptors.Following this initial culling, the selected genes were cross-referenced against an existing set of orthologous sets. The sixth orthogroup was chosen in particular for this investigation.

After the initial round of culling, the gene list underwent still another round of scrutiny. To choose which genes to keep, we looked at their expression levels and only kept those with pct.exp > 25% and avg.exp.scaled > 0.5. Only genes with high average expression levels across cells and stable expression patterns were kept for further study according to these criteria.

Following that, the purpose of the study was to evaluate whether or not the dataset contained any serotonin or dopamine receptor homologs. Using fig tree software, researchers were able to identify ant homologues of the drosophila dopamine and serotonin receptors.

DROSOPHILA	ANT
5-HT1 A 5-HT1 B	Pharaoh-ant-03257
5-HT7	Pharaoh-ant-03226
5-HT2 A	Pharaoh-ant-04214

Table 11. The serotonin Receptors of Monomorium Pharonis that is Homologous to serotonin Receptor of Drosophila

DROSOPHILIA	ANT
DOP 1R1	Pharaoh-ant-07697
DOP 1R2	Pharaoh-ant-03879
DOP EcR	Pharaoh-ant-17390
DOP 2R	Pharaoh-ant-00663

Table 12 The Dopamine receptor of Monomorium Pharonis That is homologous to Dopamine receptor of Drosophila.

Later, a dot plot analysis was carried out in order to investigate the expression patterns of serotonin and dopamine receptors throughout the various cell clusters that had been discovered. We are able to identify certain cell types or subpopulations that express these receptors

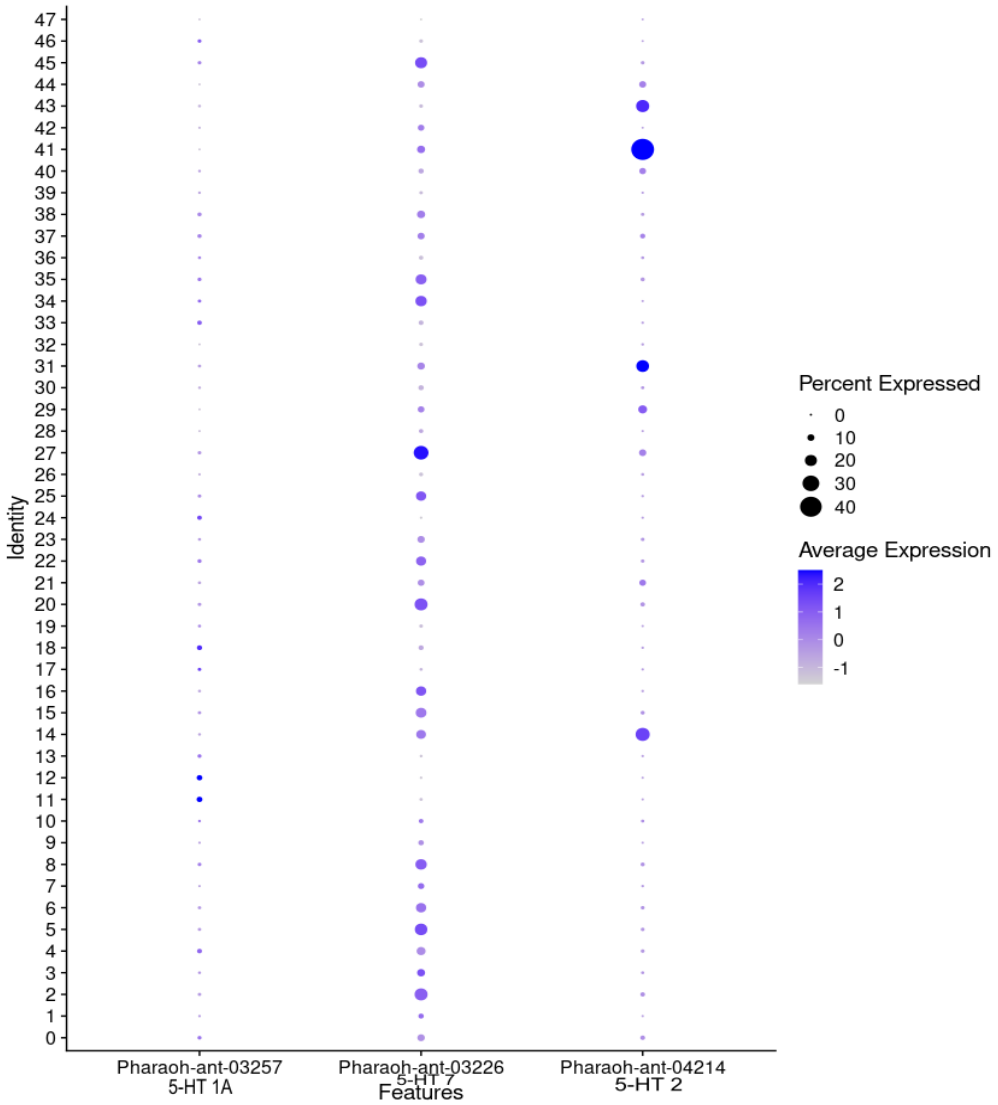


Figure 15 The serotonin receptor presence in the cluster of Monomorium Pharaonis

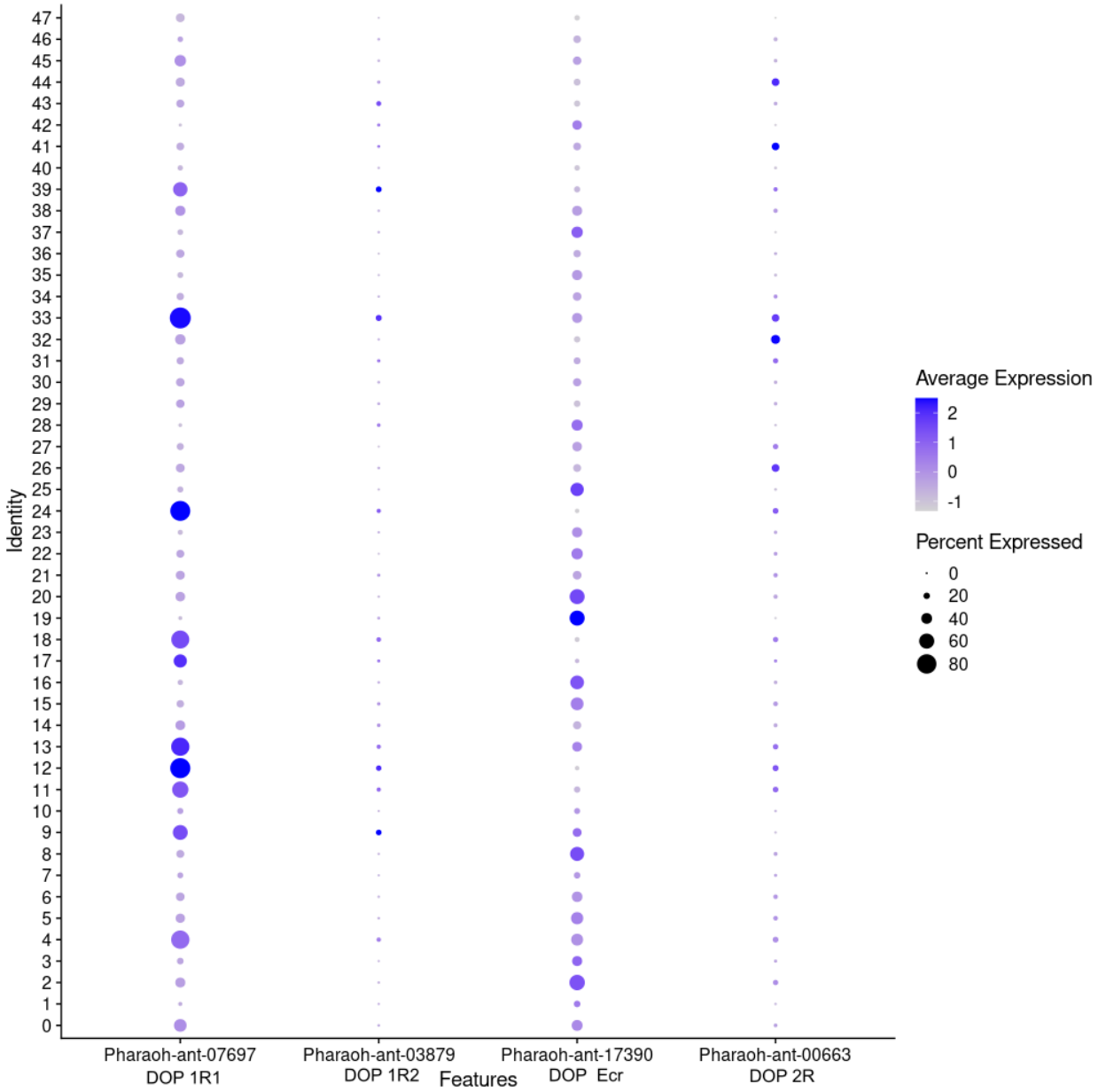


Figure 16 The Dopamine receptor presence in the cluster of Monomorium Pharaonis

3 Results

3.1 Analysis of the Serotonin and Dopamine Receptor in cluster across speceis

The exploration of receptor expression in individual clusters relied heavily on the collecting and processing of dot plot data for serotonin and dopamine receptors from three different species. To ensure that only relevant data was used for subsequent analysis, we filtered the data using criteria such as pct.exp > 25% and avg.exp.scaled > 0.5 to produce a group of receptors with substantially greater expression levels.

With this refined data set, we created bar charts for each species, showing how widely spread serotonin and dopamine receptor expression is. Each cluster is represented by its own bar in these bar graphs, and the bar's height or intensity reflects the relative percentage of receptor expression in that cluster.

Potentially useful information on the role of these neurotransmitter receptors in species-specific behaviours and physiological processes could be revealed by identifying clusters that display diverse expression profiles for serotonin and dopamine receptors in various organisms and further investigating them.

3.2 EXPRESSION PLOT

In order to better understand how neurotransmitter signalling works, we analysed the relationships between serotonin and dopamine receptors on the common cell types in Drosophila, Aedes, and Monomorium Pharaonis.

The cell type common in our data set is monoaminergic neurons,kenyon cells,glia cells,olfactory projectile neuron.We looked for the expression of the serotonin and dopamine receptor on these particular cell types in three species.Only the significant expression was marked as presence i.e with an expression percentage of more than 25% or else marked as absent.This allowed us to look for only the significant expression of the receptor.

RECEPTOR	DROSOPHILIA				AEDES				ANT			
	Monoaminergic neuron	KC	glia	PN	Monoaminergic neuron	KC	glia	PN	Monoaminergic neurons	KC	glia	PN
5HT1A	❌	❌	❌	❌	❌	✅	❌	❌	❌	❌	❌	❌
5HT1B	❌	✅	❌	✅	❌	❌	❌	❌	❌	❌	❌	❌

5HT7	❌	❌	❌	❌	❌	✅	✅	❌	✅	❌	❌	❌
5HT2	❌	❌	❌	❌	❌	❌	❌	✅	❌	❌	❌	❌
DOP 1R1	✅	✅	✅	❌	❌	✅	✅	❌	❌	✅	❌	❌

DOP 1R2	❌	✅	❌	❌	❌	✅	❌	✅	❌	❌	❌	❌
DOP EcR	❌	❌	❌	❌	❌	✅	❌	✅	❌	❌	❌	❌
DOP 2R	✅	❌	❌	✅	❌	✅	❌	✅	❌	❌	❌	❌

Figure 17 The expression plot of the common cluster in three of the species.The four cluster thet is common in three species was selected and the expression of the serotonin or dopamine receptor is marked.The ❌ indicated the expression of

the specific receptor is not in that particular cluster and ✅ indicated that the particular receptor is expressed in the particular cluster.

Monoaminergic cells

Surprisingly, it was discovered that the monoaminergic cluster in Drosophila Brain does not have any serotonin receptors expression at all. Nevertheless, it has two dopamine receptors expressed, more particularly Dop 1R1 and Dop 2R respectively. This would imply that serotonin signalling is absent in this particular cluster of Drosophila, although dopamine signalling is active across both of these types of receptors.

In Aedes, unlike in Drosophila, neither serotonin nor dopamine receptors were found in the monoaminergic cluster. This indicates that the Aedes mosquito's monoaminergic cells may not respond to serotonin or dopamine signalling, or the expression levels of these receptors may be extremely low or nonexistent in this cluster.

Ants' monoaminergic cluster had solely the 5-HT7 receptor, suggesting that this group of cells is very sensitive to serotonin signalling. However, no dopamine receptors were detected in our dataset, showing that dopamine signalling is inactive in the ant's monoaminergic cells.

Kenyon cells

There is only one serotonin receptor known to be expressed by KC cells in Drosophila, and that is 5-HT 1B. In addition, Dop 1R1 and Dop 1R2 dopamine receptors are found in these cells. Fruit fly behaviour and learning rely on serotonin and dopamine signalling, and the presence of these receptors in KC neurons shows these neurons mediate responses to these signals.

Varying receptors are expressed at varying levels on KC cells in Aedes mosquitoes. These cells express a single serotonin receptor, 5-HT 1A, suggesting a potential for a response to serotonin signalling that is unique from that seen in Drosophila. All four types of dopamine receptors (Dop 1R1, Dop 1R2, Dop EcR, and Dop 2R) are expressed by Aedes KC cells. Since these cells contain several dopamine receptors, it is likely that they are responsive to many dopamine signalling pathways and may be involved in controlling a wide variety of mosquito behaviours and sensory processes.

In contrast, Dop 1R1 is the only dopamine receptor expressed by KC cells in the ant dataset, whereas no serotonin receptors are present. When compared to Drosophila and Aedes, this receptor expression pattern in ant KC cells suggests a possible variation in the control of dopamine signalling.

Glia cells

Glia cells in Drosophila only have one dopamine receptor, called Dop 1R1. This raises the possibility that glia cells in Drosophila serve a distinct function in conjunction with dopamine signalling. The absence of serotonin receptors in these glia cells suggests that serotonin signalling does not play a direct role in glial functions in the fruit fly.

Aedes mosquito glia cells, on the other hand, express receptors in a distinct way. 5-HT7 serotonin receptor and Dop 1R1 dopamine receptor are both expressed. This provides support for the idea that glial activity in Aedes is modulated by both serotonin and dopamine signalling. The receptors' presence in glia cells indicates that they may play a more active role in neurotransmitter signalling and may exert influence over the mosquito's neural activity and behaviours.

No dopamine or serotonin receptors were found on glia cells in the ant sample. This suggests that glia cells in ants differ from their Drosophila and Aedes counterparts at the molecular level and in terms of their signalling methods.

Projectile neuron

A single 5-HT 1B serotonin receptor and a single Dop 2R dopamine receptor are expressed in Drosophila projectile neurons. Involvement in integrating and processing information from both the serotonin and dopamine systems is suggested by the presence of receptors for both neurotransmitters in these neurons. The projectile neurons in Drosophila may be regulated in part by the interaction between serotonin and dopamine signalling.

Projectile neurons of Aedes mosquitoes have a unique receptor expression pattern. They are sensitive to serotonin signalling because they express the serotonin receptor 5-HT 2. Also, these neurons express three different types of dopamine receptors: Dop 1R2, Dop EcR, and Dop 1R2. Multiple dopamine receptors indicate that Aedes's projectile neurons may be responsive to distinct dopamine signalling pathways, which may in turn control a wide variety of mosquito behaviours and sensory processes.

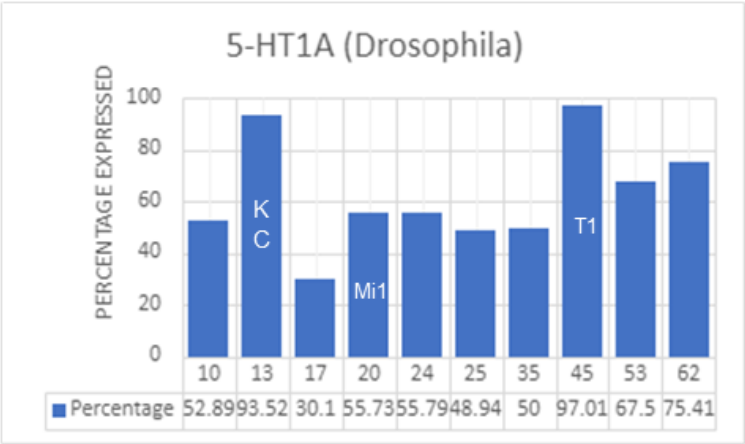
By contrast, the projectile neurons in the ant sample show no evidence of expressing dopamine or serotonin receptors. Compared to Drosophila and Aedes, ant projectile neurons may have different molecular features and signalling methods, as suggested by this result.

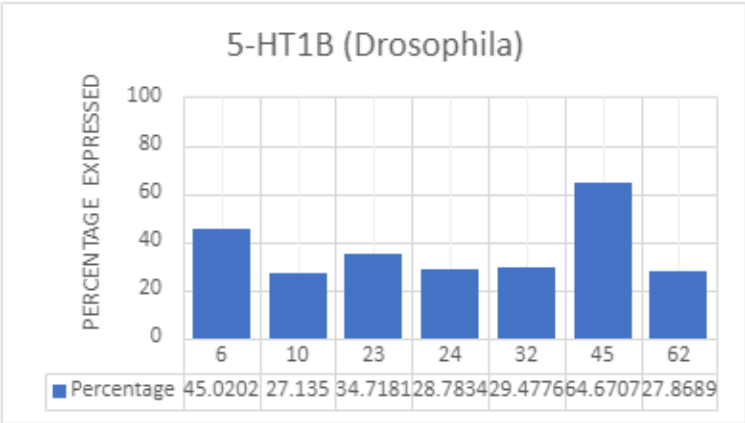
3.3 Receptor specific expression analysis

Next we looked for the specific expression pattern of the receptor in various cell types

Serotonin Receptor 1 (5-HT 1)

We first went on to examine the expression level of 5HT1A specifically in all these clusters in Drosophila. The figure below illustrates the percentage expression result of receptor 5-HT1A in Drosophila after exposure to serotonin hormone in different clusters (x-axis) that represent different cell types.


5-HT1A (Drosophila)
PERCENTAGE EXPRESSED
100
80
60
40
20
0
K
C
Mi1
T1
10 13 17 20 24 25 35 45 53 62
Percentage 52.89 93.52 30.1 55.73 55.79 48.94 50 97.01 67.5 75.41


5-HT1B (Drosophila)
PERCENTAGE EXPRESSED
100
80
60
40
20
0
6 10 23 24 32 45 62
Percentage 45.0202 27.135 34.718 28.783 29.477 64.670 27.8689

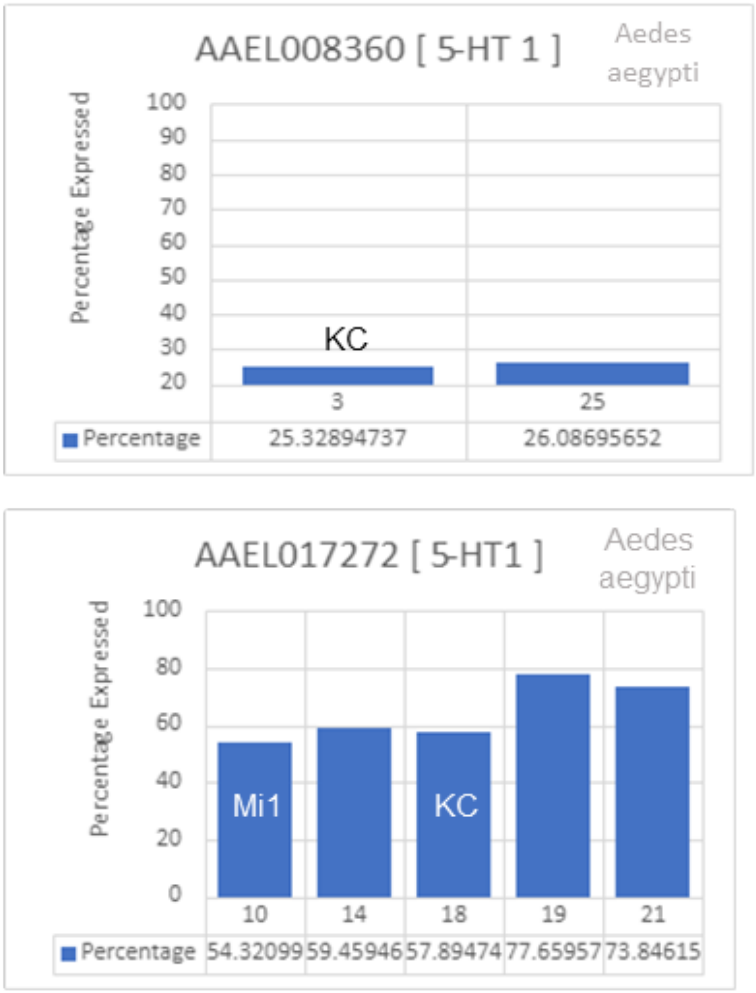


Figure 18 The bar chart showing the expression percentage of the 5-HT 1 receptor across Drosophila and Aedes aegypti.The number on bottom denotes the cluster number and the text in the chart shows the cell type of that particular cluster

It was observed in total of 10 clusters were identified to show highest visible gene expression levels for 5HT1A in Drosophila upon serotonin stimulus. Among them cluster 45 showed the highest gene expression for 5HT-1A receptor around 97 percent followed by cluster 13 with an expression level of 93.5 percent. Furthermore, from the cell annotation we know that cluster 45 belongs to the cell type T1 and the cluster 13

represents the alpha/beta KC cells thus giving us insight into what type of cells the receptor gets activated when exposed to serotonin.

For 5HT1-B, Drosophila showed highest visible gene expression levels in 7 clusters with the highest of 62% upon serotonin exposure again in cluster 45 followed by cluster 6 (45%), which was annotated as cell type Gamma KC cells and cluster 23 (34 percent) which also a type of KC cells

In case of Aedes aegypti there were two homologues of 5-HT 1 receptor AAEL008360 and AAEL017272. The data reveal that the highest visible expression levels of the AAEL008360 5-HT 1 receptor were observed in two distinct clusters, namely clusters 3 and 25. However, when compared to Drosophila, the overall expression of AAEL008360 in Aedes aegypti was lower. Cluster 3 exhibited just over 25% expression, while cluster 25 showed a slightly higher level at 26%. Interestingly, through cell annotation, cluster 3 was identified as comprising KC (Kenyon Cells) cells in the Aedes mosquito.

Different expression patterns were discovered for AAEL017272. The highest level of expression was found in cluster 19, at 77.6 percent, followed by cluster 21 (73.8 percent), cluster 18 (57.8 percent), cluster 14 (59.45 percent), and cluster 10 (54.4 percent). Interestingly, Mi1 cells were found in cluster 10, while KC cells were found in cluster 18.

Aedes aegypti mosquitoes, like many other insects, have Mi1 cells in their brains. The mosquito's antennae are its primary olfactory organs, and these cells play a critical role in interpreting the sensory information they receive. Mi1 cells are involved in the detection and interpretation of odours, and they are found in the olfactory system.

The presence of the 5-HT 1 receptor in Mi1 cells suggests that it may play a function in regulating the processing of olfactory signals by these neurons. The receptor may have a role in mosquito olfactory perception and response, which in turn may affect behaviours including host-seeking and oviposition site selection.

The fact that cluster 18 is made up of KC cells and has a substantial expression level of the AAEL017272 receptor lends credence to the hypothesis that this receptor is

involved in the regulation of the processes of learning and memory in mosquitoes. The presence of the receptor in KC cells suggests that it may be involved in the storage of memories connected to sensory events. This may have an effect on the mosquito's capacity to associate particular stimuli with specific behaviours or results.

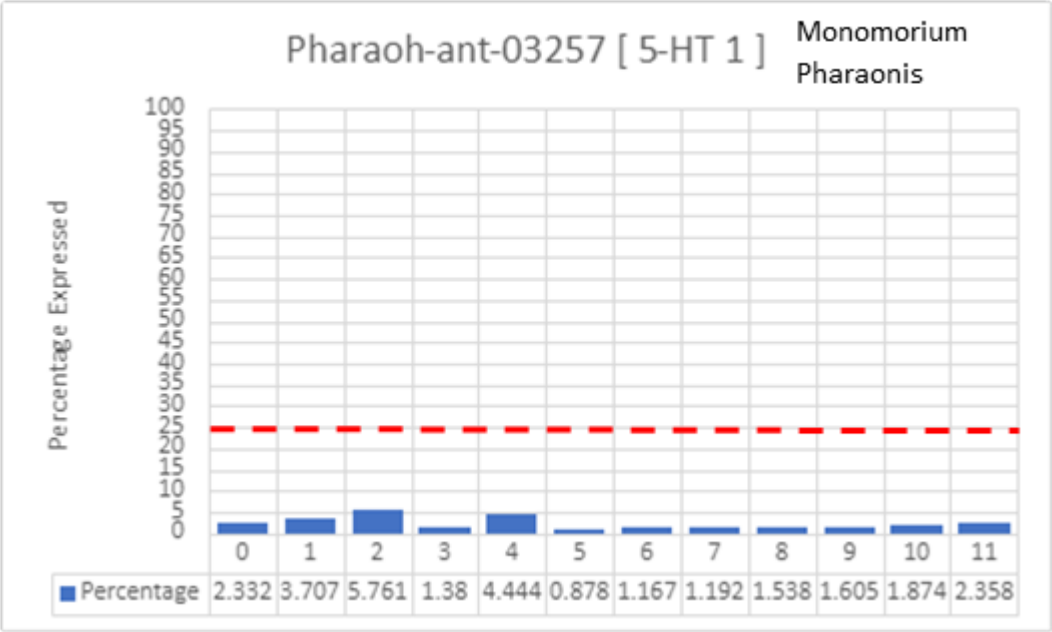


Figure 19 The Graph of the 5-HT 1 receptor homolouge in Monomorium Phaonis.The Red Dot line is the threshold that we set for the significant receptor expression.

On the other hand, the ant data set does not contain any substantial expression of the 5-HT1 receptor. We conducted additional research into this topic to establish that the lack of expression is not the result of poor clustering.We created a feature map of the 5-HT homologue of the ant, and there were no major cells that expressed the receptor in the ant.

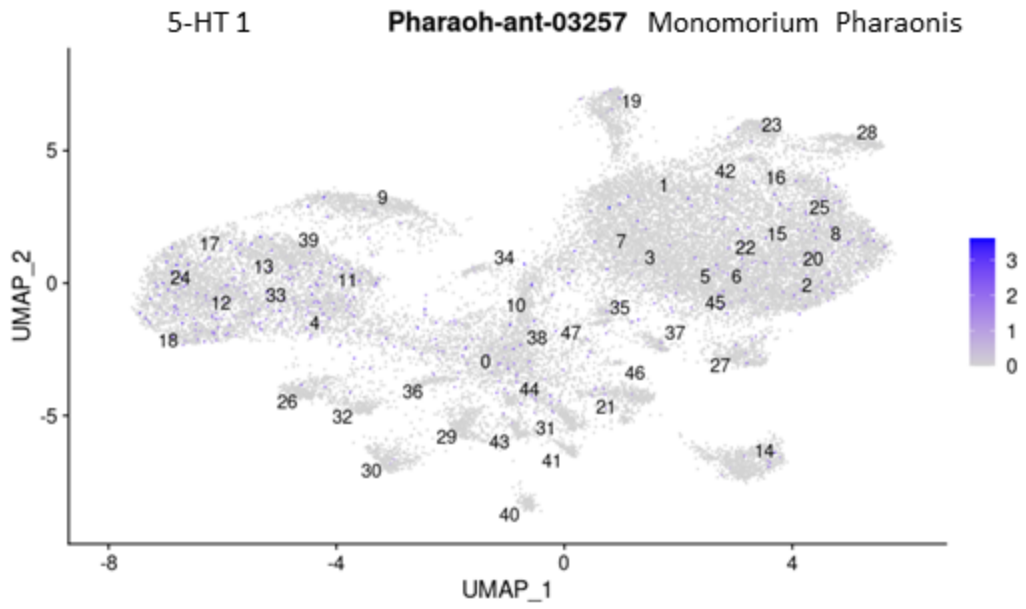
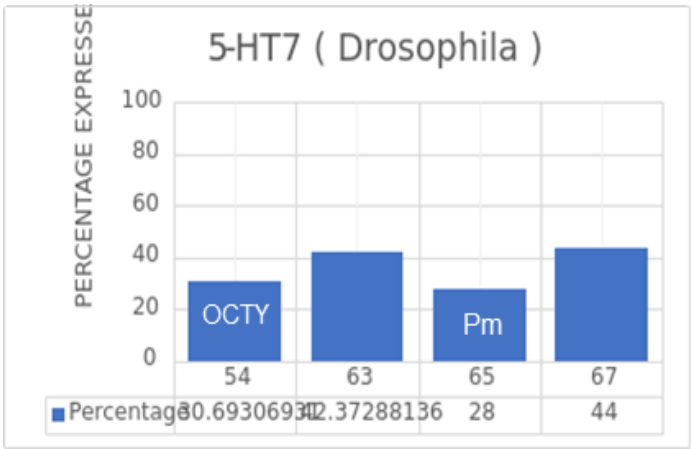
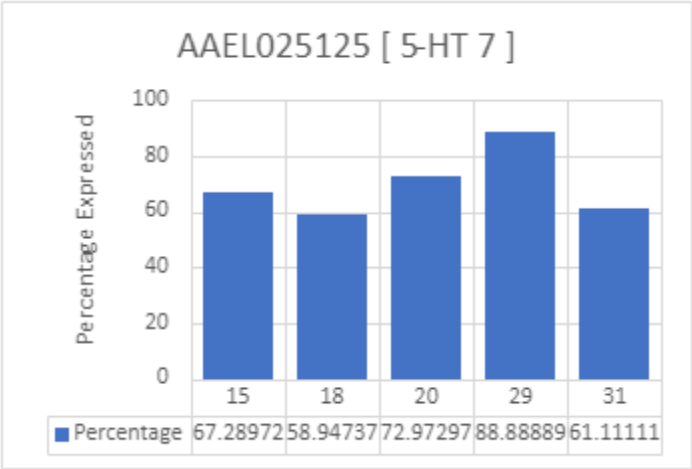
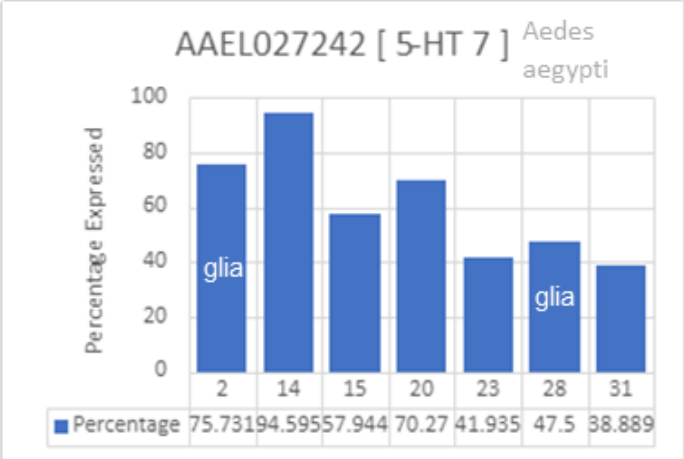


Figure 20 The feature plot of the 5-HT 1 receptor homolouge of Mnomorium Pharaonis.The recptor seems to present in very few cells.

Serotonin Receptor 2 (5-HT 7)







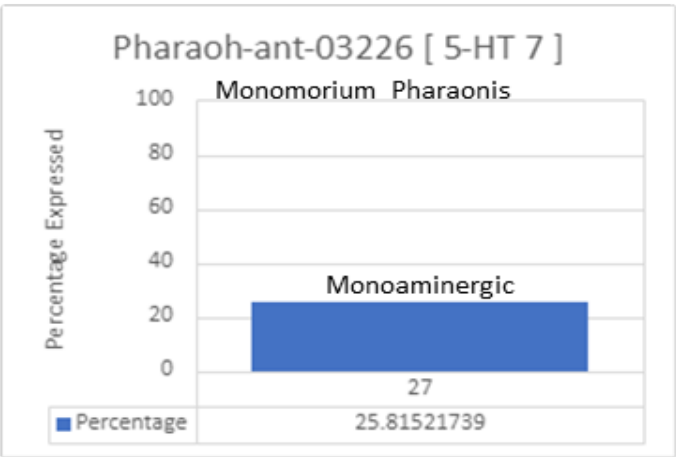


Figure 21 The bar chart showing the expression percentage of the 5-HT 7 receptor across Drosophila,Aedes aegypti and Monomorium Pharaonis.The number on bottom denotes the cluster number and the text in the chart shows the cell type of that particular cluster

The 5-HT 7 receptor is shown to be differentially expressed in Drosophila across four clusters as indicated by the bar graph. The highest level of expression (44%) is found in cluster 67, followed by clusters (63), (54), and (65). Even though the receptor's average expression level isn't particularly high, there is a wide range of values within each of the clusters that have been found.The cluster 65 was annotated as Pm3 cells but the receptor expression was comparatively low of 28 percent.

In ants, the receptor is mostly expressed in cluster 27, with a percentage of expression of 25%. This group of genes is labelled as "Monoaminergic," suggesting that they may play a function in controlling the activity of monoamine neurotransmitters in the ant species under investigation.

There are two homologues of the 5-HT 7 receptor in Aedes mosquitoes, AAEL025125 and AAEL027242. Clusters 15, 20, and 31 show high levels of expression for both homologues. These groups of genes almost certainly serve crucial functions in the physiological processes of the 5-HT 7 receptor in the Aedes mosquito.

Serotonin Receptor 3 (5-HT 2A)

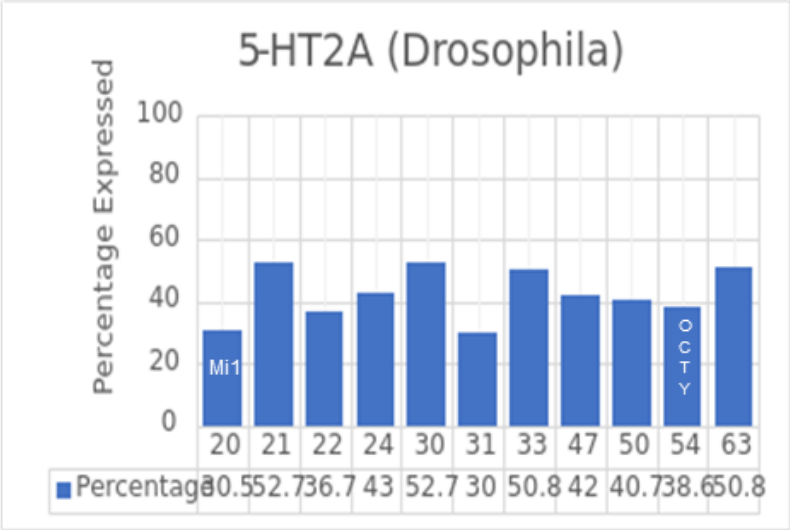
The 5-HT 2A receptor is expressed in 11 distinct regions in the fruit fly, Drosophila. At levels of expression more than 50%, these groups 21, 33, and 63 stand out. These groups of genes may play a significant role in modulating responses to serotonin signalling, hence affecting the wide range of behaviours and physiological processes that serotonin controls.

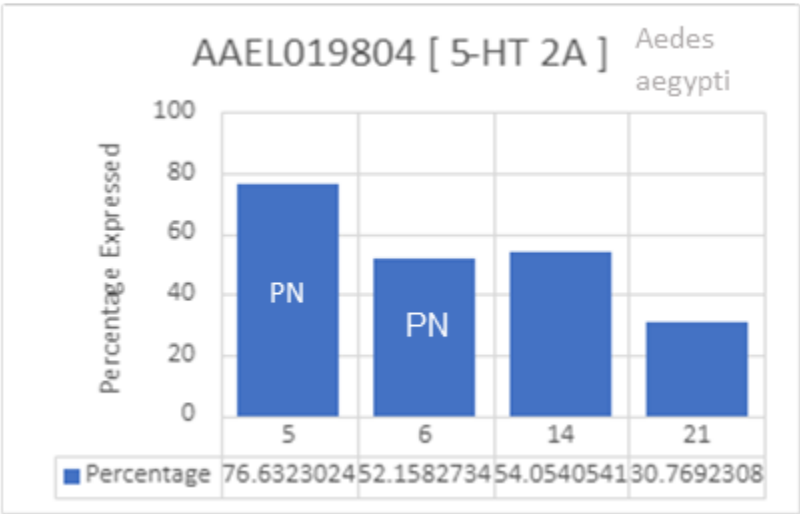
The 5-HT 2A receptor is highly expressed in clusters 54 (annotated as OCTY) and 20 (annotated as Mi1), with both clusters showing percentages of expression above 30%. These neuronal populations may be more responsive to serotonin signalling, suggesting a role in the fine-tuning of some Drosophila behaviours and senses.

The receptor is significantly expressed in three different clusters in Aedes mosquitoes, all of which have percentages higher than fifty percent. The fact that two of these clusters are associated with projectile neurons demonstrates the tight relationship that exists between the 5-HT 2A receptor and the projectile neuron. The high expression of

the receptor in the projectile neurons of Aedes mosquitoes suggests that it may have a role in controlling the motor behaviours and sensory functions of these insects.

Cluster 41 in the ant dataset had the highest expression of the 5-HT 2A receptor, at 42%. This expression level is lower than in some Drosophila and Aedes clusters, but it nevertheless indicates the presence of the receptor in certain neurons of ants. More research is needed to determine the functional importance of this receptor's expression in ant neuron.





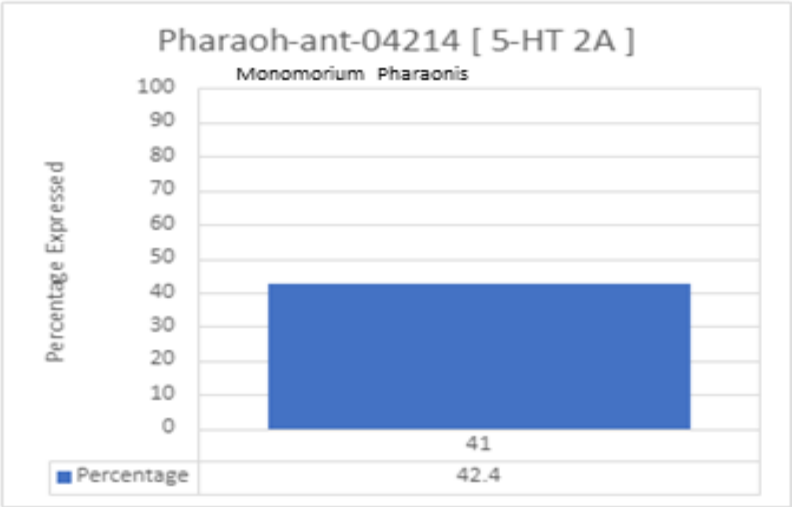
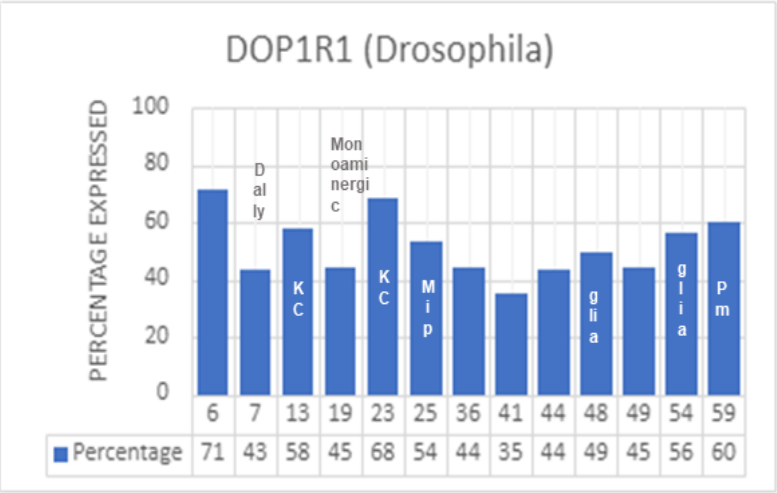


Figure 22 The bar chart showing the expression percentage of the 5-HT 1 2A receptor across Drosophila, Aedes aegypti and Monomorium Pharonis.The number on bottom denotes the cluster number and the text in the chart shows the cell type of that particular cluster.

Dopamine Receptor 1 (DOP 1R1)



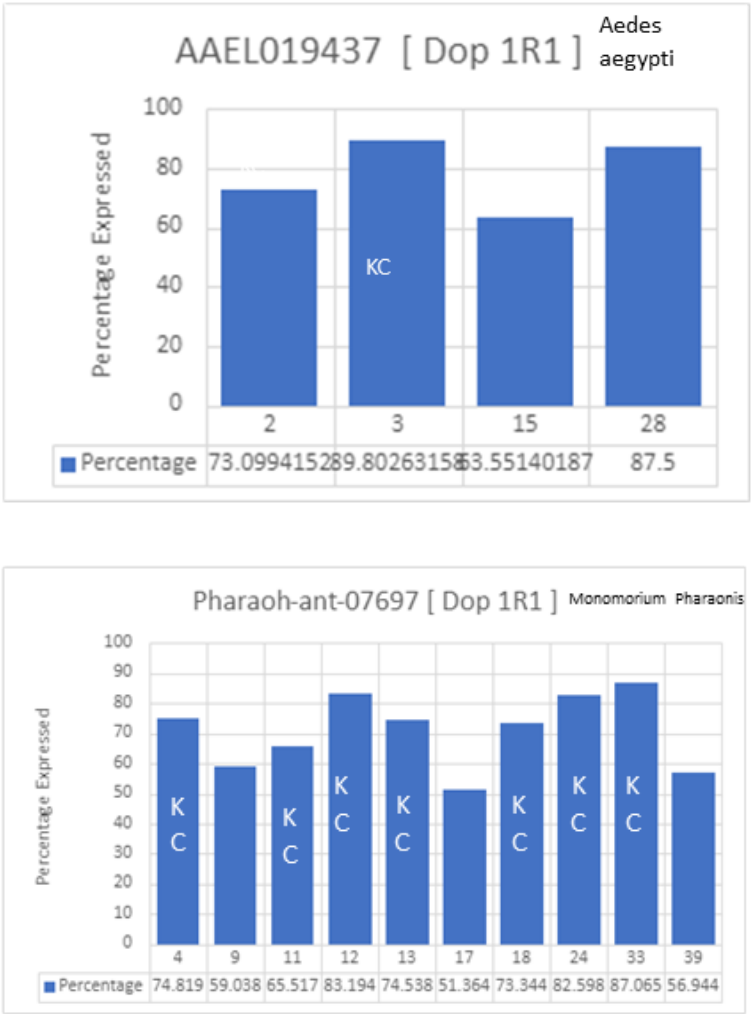


Figure 23 The bar chart showing the expression percentage of the DOP 1R1 receptor across Drosophila, Aedes aegypti and Monomorium Pharaonis.The number on bottom denotes the cluster number and the text in the chart shows the cell type of that particular cluster

The Dop 1R1 receptor is highly expressed in several populations of Drosophila. The KC cells, on average, show the highest expression, at around 60%. This points to a critical function for the Dop 1R1 receptor in these cells, which are well-known for their part in olfactory learning and memory. An average expression level of 60% is also observed in Pm cells, suggesting a potential role in the regulation of locomotor behaviours and other activities of these neurons. There may be interactions between the Dop 1R1 receptor and the octopaminergic and tyraminergic signalling systems, as shown by the OCTY

cells' expression percentage of 54.Additional cell types expressing the Dop 1R1 receptor in Drosophila are indicated by the expression percentage of 45 in the monoaminergic cluster.

The KC cells of ants express the Dop 1R1 receptor at a rather high level, often around 75%. This suggests that the receptor may play a key function in these cells, which are essential to ant learning and memory. Six distinct KC cell clusters express the Dop 1R1 receptor, suggesting that this receptor may play a significant role in modulating olfactory and memory-related brain activities.

Dop 1R1 receptor is expressed at a rate of 75% on average in glia cells of Aedes mosquitoes. Glia cells are neuronal support cells that play essential functions in maintaining neural connectivity and cellular balance. Given the high level of receptor expression in these cells, one can hypothesise that it has a role in regulating glial activities, hence potentially impacting neuronal signalling and behaviour in Aedes.

Dopamine Receptor 2 (DOP 1R2)

Dop 1R2 receptor is highly expressed in all three clusters of Kenyon cells in Drosophila, with expression percentages that are greater than 80 percent in each cluster. This suggests that the receptor is likely to play a crucial function in these cells, which are well-known for their participation in the processes of olfactory learning and memory. The high levels of expression in Drosophila suggest that the Dop 1R2 receptor may be a crucial participant in controlling the synaptic plasticity and brain circuits that are the basis for the generation of olfactory memories in this species.

Moreover, the OCTY cells express the Dop 1R2 receptor at a rate greater than 60%. The presence of the receptor in these cells is consistent with its association with octopaminergic and tyraminergic signalling, suggesting that the receptor may interact with these neurotransmitter systems to regulate a variety of fruit fly physiological processes and behaviours.

Each of the three groups of Kenyon cells in an ant's brain expresses the Dop 1R2 receptor at a rate greater than 80%. This demonstrates how crucial the receptor is for

ants' olfactory learning and memory. The high levels of expression of the Dop 1R2 receptor in Kenyon cells indicate that this receptor plays a critical role in these neurons across species.

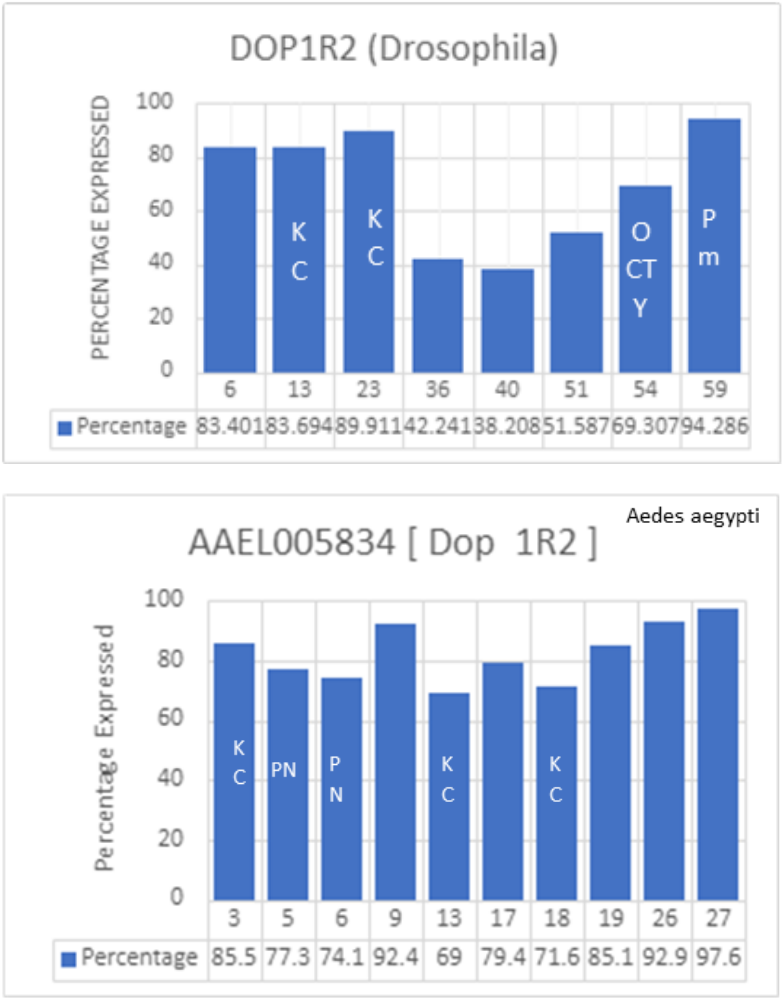


Figure 24 The bar chart showing the expression percentage of the DOP 1R2 receptor across Drosophila, Aedes aegypti and Monomorium Pharaonis.The number on bottom denotes the cluster number and the text in the chart shows the cell type of that particular cluster

The Dop 1R2 homologue in Monomorium pharonis was not expressed in our cluster, which could be the result of our decision to only consider clusters with expression levels

of at least 25% or may be due to poor clustering.so we decided to make a feature plot to visualise the receptor expression across the cluster.But the feature plot analysis did not provide any significant result.

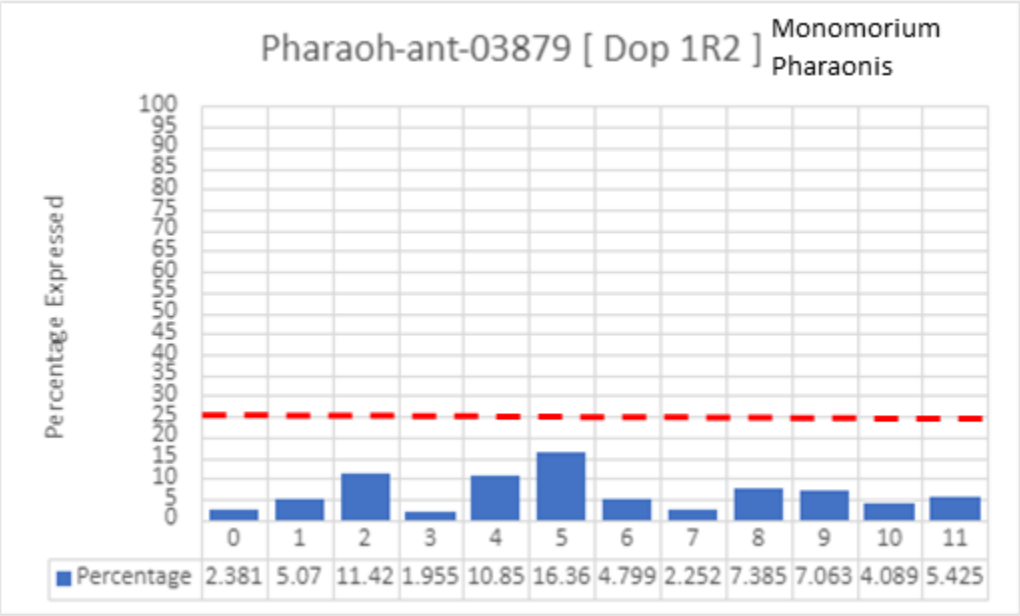


Figure 25 The Dop 1R1 receptor in Monomorium Pharaonis .The red Dot line is the Threshold we set for the significant receptor expression.since the receptor did not pass the expression percentage we considered them to be not significant.

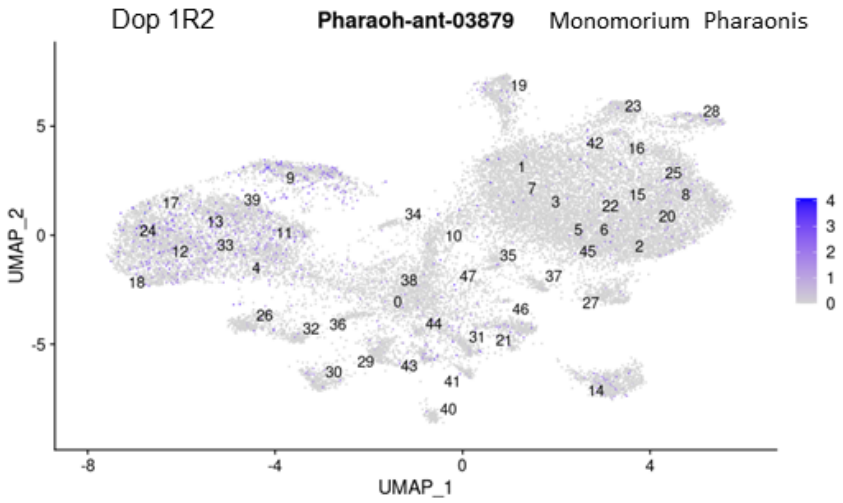


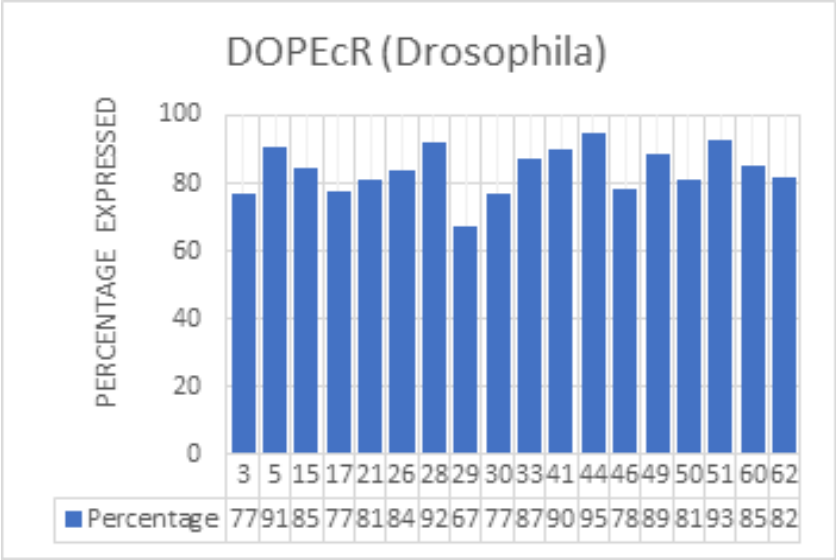
Figure 26 The feature plot of the Dop1R2 receptor in Monomorium Pharaonis .It is clear thta there is very few cells expressing the receptor.

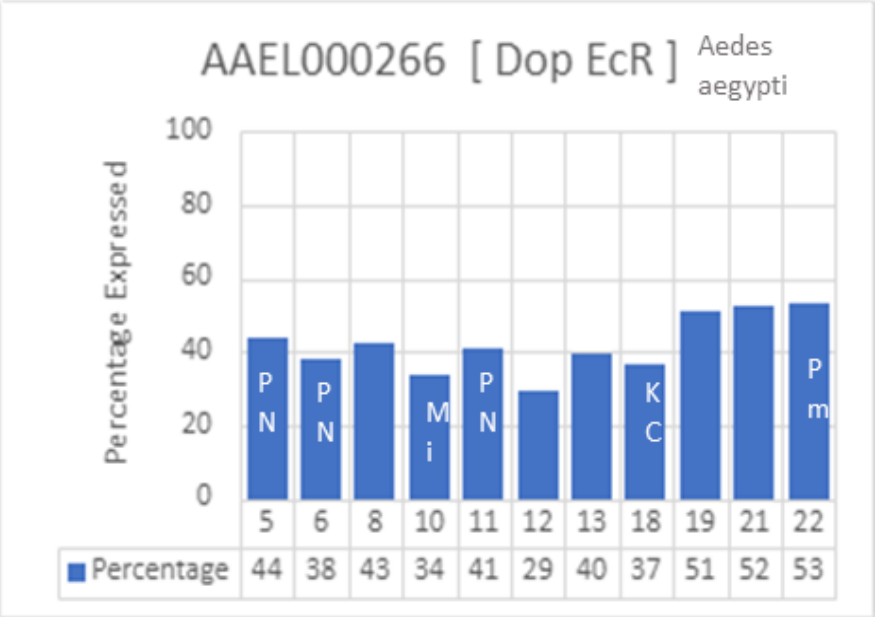
Dopamine Receptor 3 (DOP EcR)

Dop EcR receptor is expressed at quite high levels throughout Drosophila, with an average expression of roughly 80% across 18 distinct clusters. However, the majority of these clusters have unidentified cell types. The abundance of Dop EcR receptor expression in fruit flies indicates that it likely plays important functions in a wide range of brain circuits and behaviours. Additional studies are required to determine which cell types express the receptor and to decipher its function.

The Dop EcR receptor is found in 11 clusters in ants, with an average expression of more than 40 percent in each group. Cluster 19 is made up of T4/T5 cells in the optic lobe, and its rate of expression is 60. Since the Optic lobe T4/T5 cells are known for their roles in vision, this suggests that the receptor may be involved in controlling how ants process vision.

The Dop EcR receptor is expressed in 11 distinct clusters in Aedes mosquitoes, with a median expression level more than 40 percent. The average expression level is roughly 40% in projectile neurons and Kenyon cells. The receptor is also expressed at higher rates (53%) in Pm cells than in Mi1 cells (34%). Aedes mosquitoes may use this receptor to control motor behaviours, olfactory learning, and other brain functions because it is present in projectile neurons, Kenyon cells, and other cell types.


DOPEcR (Drosophila)
PERCENTAGE EXPRESSED
100
80
60
40
20
0
3 5 15 17 21 26 28 29 30 33 41 44 46 49 50 51 60 62
Percentage 77 91 85 77 81 84 92 67 77 87 90 95 78 89 81 93 85 82


AAEL000266 [Dop EcR]
Aedes aegypti
Percentage Expressed
100
80
60
40
20
0
P N
P N
M i
P N
K C
P m
5 6 8 10 11 12 13 18 19 21 22
Percentage 44 38 43 34 41 29 40 37 51 52 53

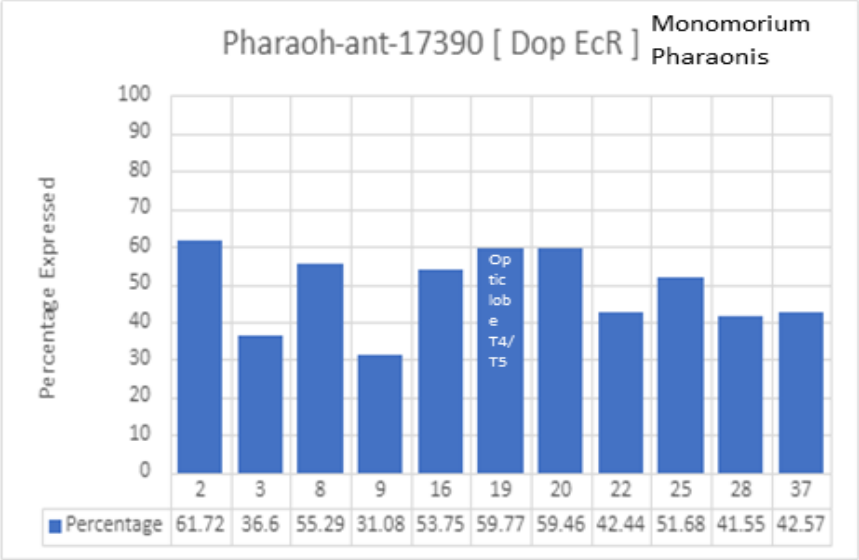
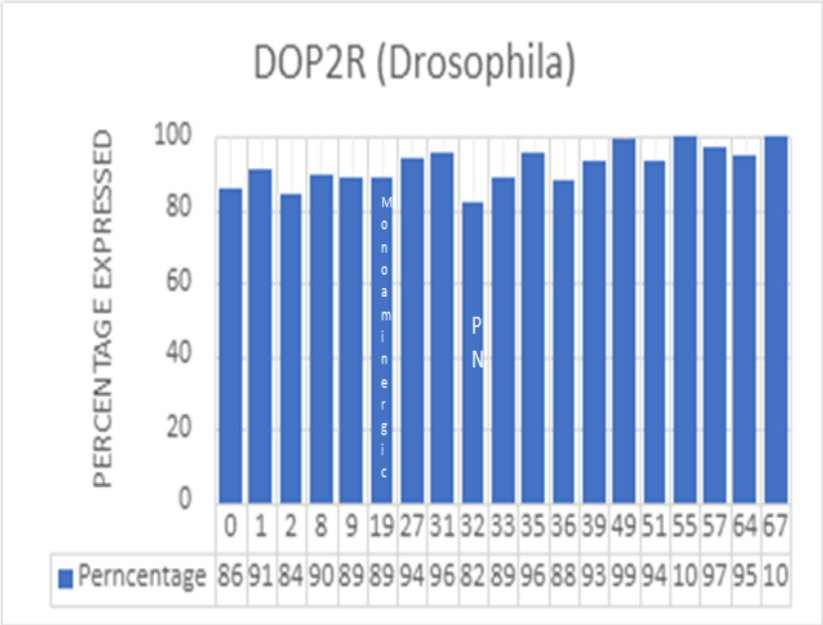
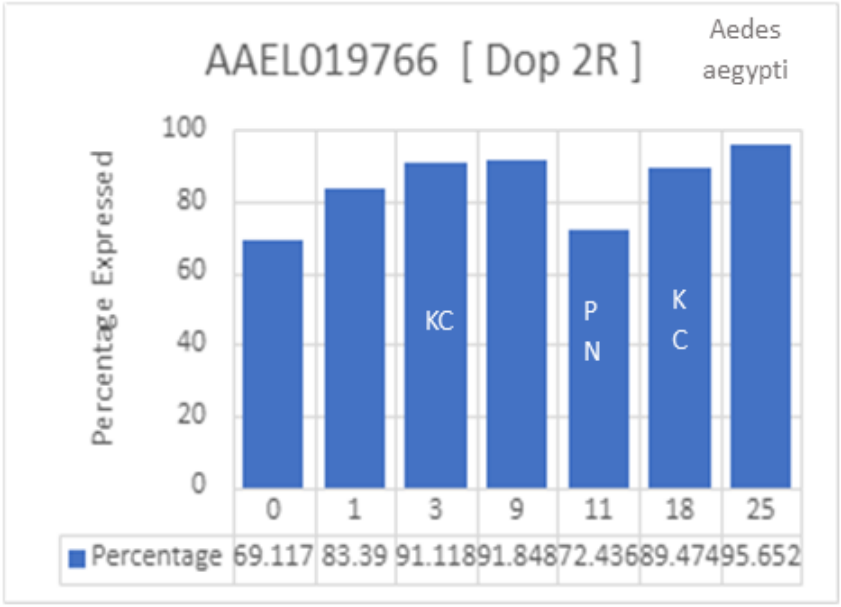


Figure 27 The bar chart showing the expression percentage of the DOP EcR receptor across Drosophila, Aedes aegypti and Monomorium Pharaonis.The number on bottom denotes the cluster number and the text in the chart shows the cell type of that particular cluster

Dopamine Receptor 4 (DOP 2R)





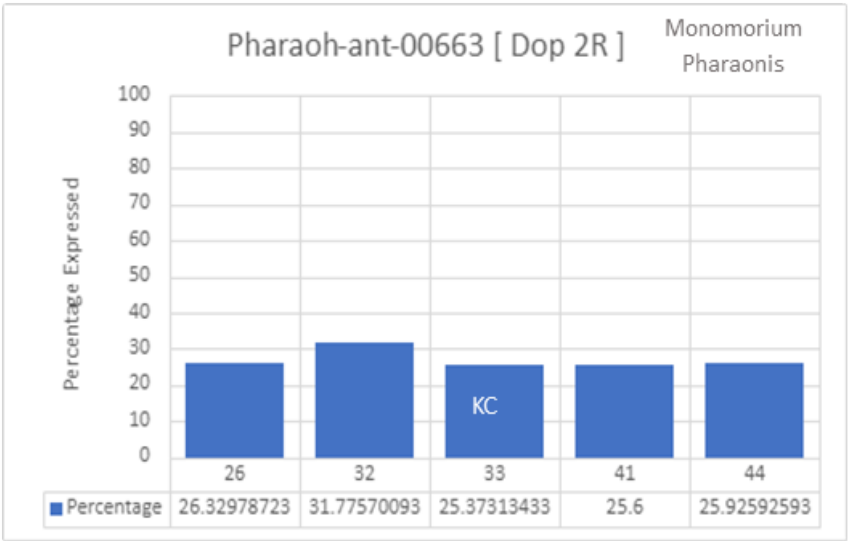


Figure 28 The bar chart showing the expression percentage of the DOP 2R receptor across Drosophila, Aedes aegypti and Monomorium Pharaonis.The number on bottom denotes the cluster number and the text in the chart shows the cell type of that particular cluster

The Dop 2R receptor is over-expressed in 19 distinct regions of Drosophila. Clusters 55 and 67 both show a 100% expression rate, meaning that every single cell in those clusters has the receptor gene. Cluster 49 is really close behind, expressing at a 99

percent level. Dop 2R receptor may play critical roles in the brain circuits and behaviobehaviors with these clusters, as evidenced by their high expression levels.

Cluster 19 is made up of monoaminergic cells and expresses Dop 2R at a rate of 89%, while cluster 32 is made up of olfactory projectile neurons and expresses Dop 2R at a rate of 82%. As the receptor is found in these clusters, it may play a role in regulating monoaminergic signaling and olfactory processing in Drosophila.

The Dop 2R receptor is expressed at a steady rate of 80 percent across seven distinct loci in Aedes mosquitoes. Cluster 11 of projectile neurons express the receptor at a rate of 72%, and cluster 18 of Kenyon cells express it at a rate of 89%. The inclusion of the dopamine 2R receptor in these groups raises the possibility that it plays a function in the regulation of motor behaviors and olfactory learning in Aedes mosquitoes.

Dop 2R receptor expression is low across the board, at around 25% in ants. Because none of the cell types that make up the clusters are known, additional research is required to determine the particular brain circuits and behaviors that are connected with the receptor in ants.

4 Discussion

Similarities in how serotonin and dopamine receptors are expressed in KC cells and glia cells between Aedes and Drosophila are interesting and show that these receptors may have the same functions in these cell types in both species.

Dop1R1 is found in the glial cells of both Aedes and Drosophila, which suggests that it might have the same function in both species. This could be linked to how glial cells in both species affect synaptic transmission.

In addition to Aedes and Drosophila, Monomorium pharaonis also makes the Dop1R1 receptor. The fact that Dop1R1 is found in Aedes, Drosophila, and Monomorium pharaonis shows that this receptor serves the same purpose in all of these species. This shared expression suggests that Dop1R1 may have a role that may have not

changed over time in controlling certain cellular processes or behaviours that are common to these insects.

Also, the fact that Dop2R is found in projectile neurons in both Aedes and Drosophila suggests that it may control motor function and movement, since projectile neurons in insects control and coordinate movement.

Overall, the fact that the same serotonin and dopamine receptors are expressed in the same cell types in both Aedes and Drosophila suggests that these receptors may have the same roles in both species to control neural processes, synaptic transmission, and behavioural responses. But more functional studies and manipulations, like receptor-specific agonist/antagonist treatments, would be needed to find out for sure what role these receptor expression patterns play in specific behaviours or physiological processes in Aedes and Drosophila. The results are a good starting point for future study that will look into how these receptors work in these types of cells and how they affect the neurobiology and behaviour of these two important insect species.

The fact that both Aedes and Drosophila have 5-HT1 receptors in their Mi1 cells adds to the idea that these insects have the same functions. The fact that Mi1 cells in both species have the 5-HT1 receptor shows that this receptor has the same role in controlling how these cells work.

The 5-HT1 receptor is a serotonin receptor, and it is known to play a part in regulating neural activity and synaptic transmission. Its presence in the Mi1 cells in the optic lobes of the brain, which are responsible for processing vision and detecting motion, shows that serotonin signalling could affect these processes in both Aedes and Drosophila.

5 Conclusion

The main goal of this study is to find the expression of the monoamine receptor, especially serotonin and dopamine across different species of insects. The scRNA-seq data from three insect species: Aedes aegypti, Monomorium pharaonis, and Drosophila was gathered. Poor-quality cells were filtered out of the data, and gene expression levels were normalized. Cells were subsequently clustered based on their gene

expression profiles using a clustering approach. Within each cluster, the marker genes correspond to distinct cell types and annotated those cell types using marker genes that were already known from the literature and public sources.

The primary goal of the study was to compare and contrast the receptor expression of serotonin and dopamine between the three species. Within each species, the presence and distribution of cells expressing serotonin and dopamine receptors were revealed by analyzing the expression levels of these receptors in specific cell types. In addition, a comparison across species was done to see if there were any conserved or species-specific differences in the expression patterns of receptors.

The research on the expression patterns of serotonin and dopamine receptors across a variety of insect species using single-cell RNA sequencing (scRNA-seq) is extremely helpful and has the potential to be applied in many different fields. To begin, it contributes significant knowledge to the field of comparative neurobiology by highlighting the similarities and differences in the expression of neurotransmitter receptors across a wide range of insect species. Neurobiology can learn a lot from understanding how neural pathways and behaviors controlled by these receptors change over time. Second, the results contribute to research on the physiology and behavior of insects, since serotonin and dopamine receptors are essential for many processes, such as movement, feeding, mating, and responses. This information can be useful in the development of methods for pest control and in the conduct of research on the relationships between insects and their hosts. Additionally, the findings can also contribute to the development of new drugs, as serotonin and dopamine receptors are possible targets for pesticides and repellents that can selectively affect insect pests while sparing non-target animals. In addition to this, it also contributes to our knowledge of human neurological diseases as this receptor plays a key role in neuron signaling. This work also makes a contribution to the field of evolutionary studies by shedding light on the patterns of receptor expression and the linkages that exist between the various species of insects. This study will serve as a groundwork for future research, directing inquiries into the particular functions that serotonin and dopamine receptors play in

various cell types. In general, the applications of this work range from fundamental science to more applied sectors.

Even though the study's findings are quite helpful, it does have some drawbacks that need to be noted. In the first place, the investigation was primarily concerned with the expression of serotonin and dopamine receptors in insects, and the scope of the investigation was restricted to just three different species: Aedes aegypti, Monomorium pharaonis, and Drosophila. As a result, the findings might not apply to other insects or higher organisms. If the scope of the study were broadened to encompass a greater variety of insect species, a better understanding of the patterns of receptor expression that have been observed throughout many evolutionary lineages may be attained. Another weakness of the study is that it did not directly evaluate the activity or function of the serotonin or dopamine receptors; rather, it focused on the mRNA expression of these receptors. This does not always correlate.Hence further investigation of the actual receptor is needed.

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